Chapter 1: Complex Numbers and Complex Functions

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September 27, 2026

Outline

1. Complex Numbers and Their Expressions

2. Operations of Complex Numbers

3. Domains in the Complex Plane

4. Complex Functions

5. Limits and Continuity of Complex Functions

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1 The Concept of Complex Numbers

In the real number system, the equation $x^2 = -1$ has no solution. To solve such equations, we introduce the imaginary unit (虚数单位) i (often denoted by j in electrical engineering), defined by $i^2 = -1$. Thus, i is a solution of $x^2 = -1$.

For any real numbers x and y, the number $z = x + iy$ is called a complex number (复数). The quantities x and y are the real part (实部) and imaginary part (虚部) of z, respectively, and are denoted by $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$.

When the coefficient of i is numerical, we usually write i after the coefficient, as in $2+3i$. When the coefficient is a variable, we often write i before it, as in $a+ib$.
If $x=0$ and $y\neq 0$, then $z=iy$ is called a purely imaginary number (纯虚数) . If $y=0$, then $z=x$, so every real number is naturally a complex number.

2 Expressions of Complex Numbers

A complex number $z = x + iy$ is uniquely determined by the ordered pair(有序数对) (x, y). Therefore, complex numbers are in one-to-one correspondence with points in a two-dimensional Cartesian coordinate system（二维直角坐标系）: the number $z = x+iy$ is represented by the point (x, y). This plane is called the complex plane(复平面), or the z-plane. The x-axis is called the real axis (实轴), and the y-axis is called the imaginary axis (虚轴).

Each complex number $z = x + iy$ determines exactly one point (x, y) in the complex plane, and vice versa. This correspondence allows us to study complex functions using geometric language and methods, and it provides a geometric foundation for many engineering applications.

In the complex plane, $z = x+iy$ also corresponds to the position vector (位置向量) $\overrightarrow{OP}$ from the origin (原点) O to the point $P = (x, y)$ (see Figure 1.1). This is the vector representation (向量表示) of z. The length of this vector is called the modulus (模), or absolute value (绝对值), of z and is denoted by

$$|z| = r = \sqrt{x^2 + y^2} \tag{1.1.1}$$

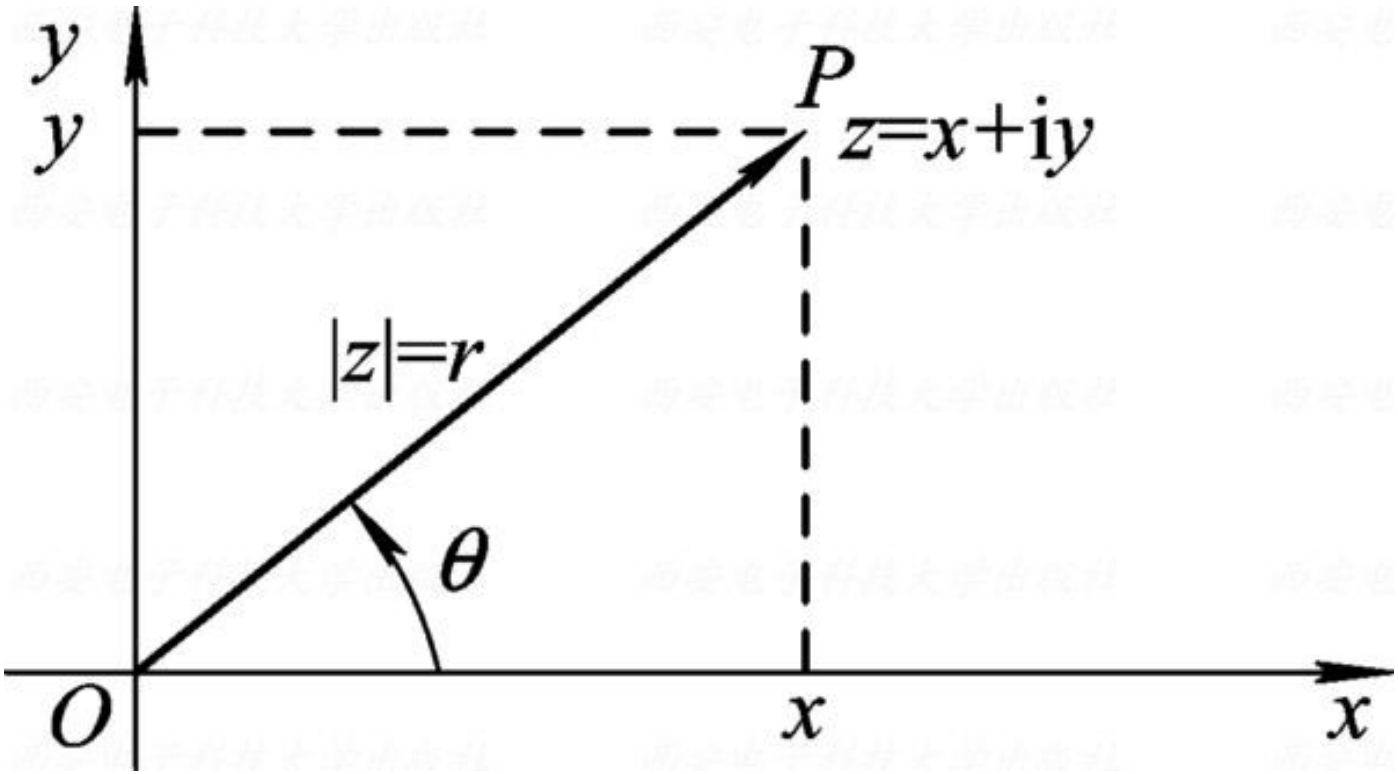


Figure: 1.1

The following inequalities are immediate:

$$|x| \le |z|\,, \quad |y| \le |z|\,, \quad |z| \le |x| + |y|\,.$$

For $z \neq 0$, an angle θ measured counterclockwise (逆时针) from the positive real axis to the vector $\overrightarrow{OP}$ is called an argument (幅角) of z. Since $x = r\cos\theta$ and $y = r\sin\theta$, we have

$$\tan\theta = \frac{y}{x} \quad (x \neq 0). \tag{1.1.2}$$

Every nonzero complex number has infinitely many arguments (see Figure 1.2). If θ_1 is one argument of z, then all its arguments are

$$\operatorname{Arg} z = \{\theta_1 + 2k\pi : k \in \mathbb{Z}\}. \tag{1.1.3}$$

Here, $\operatorname{Arg} z$ denotes the set of all arguments of z.

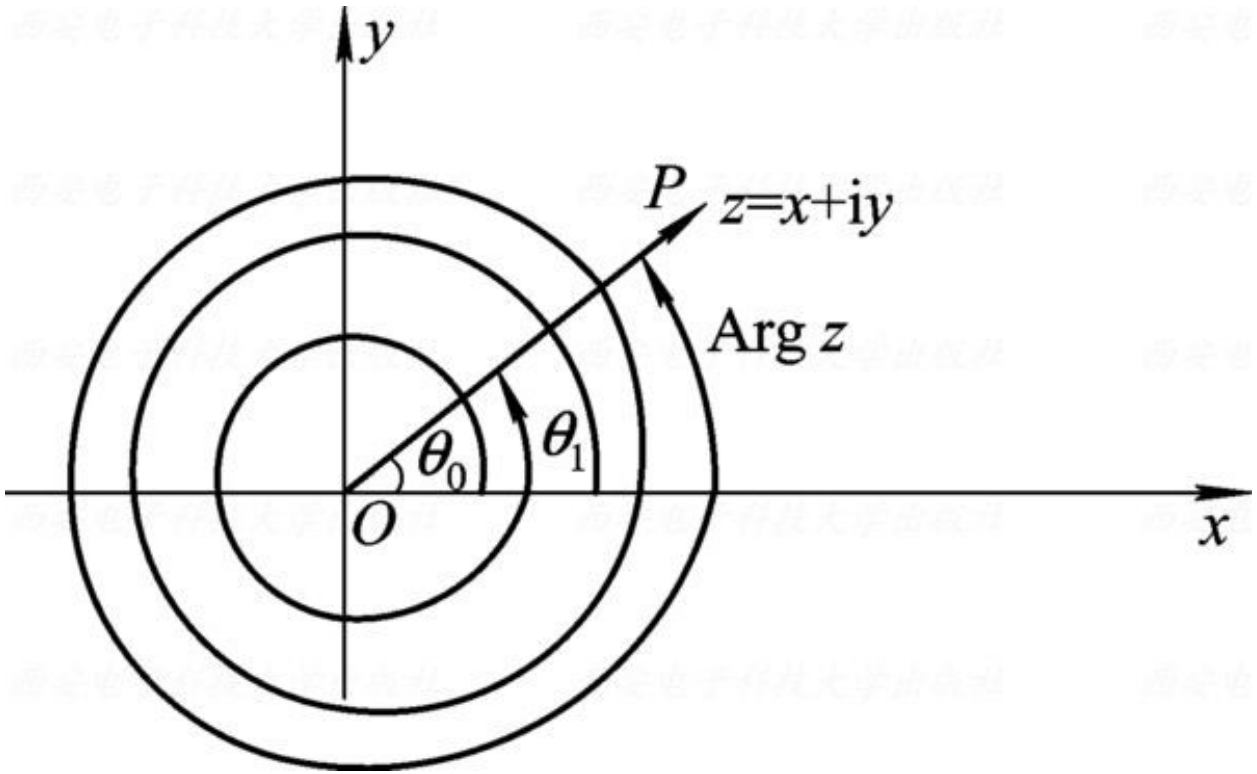


Figure: 1.2

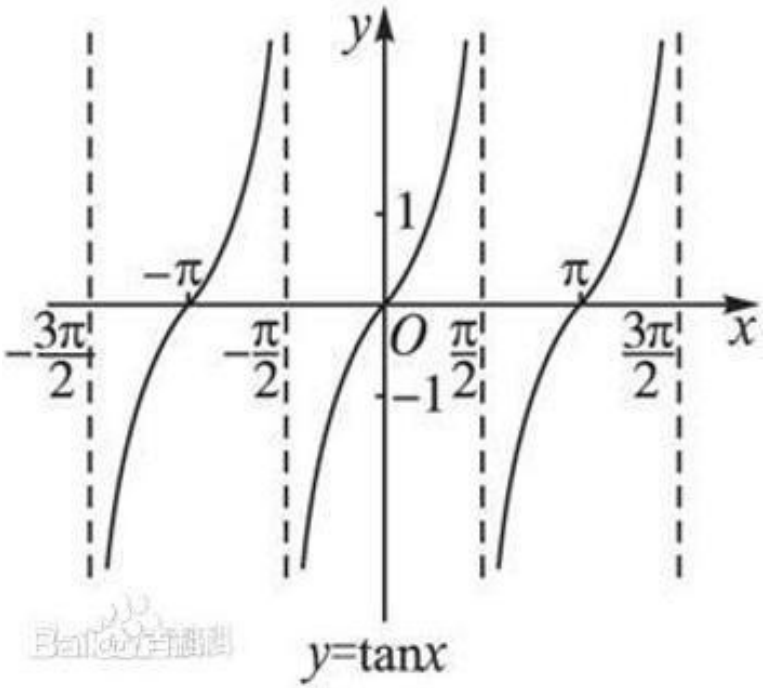
Among all arguments of z, the unique value θ_0 satisfying $-\pi < \theta_0 \leq \pi$ is called the principal argument (幅角主値) and is denoted by $\arg z = \theta_0$. Hence,

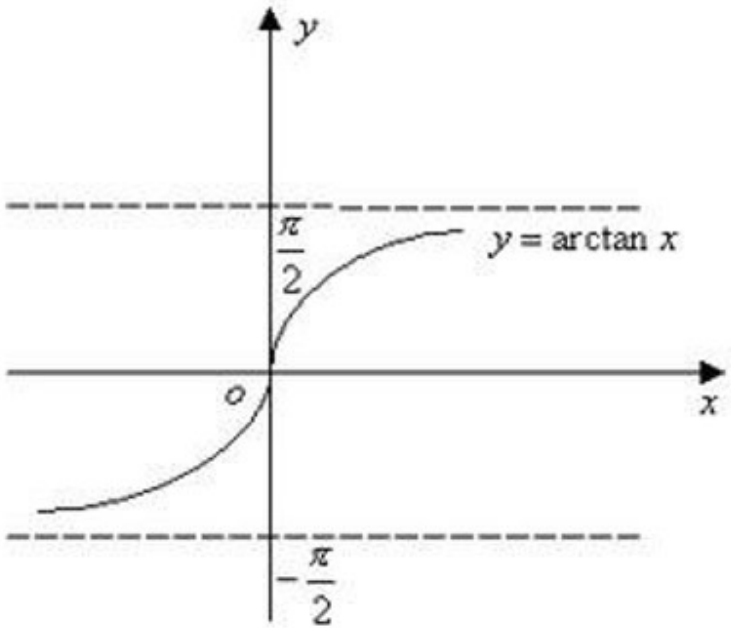
$$\operatorname{Arg} z = \{\arg z + 2k\pi : k \in \mathbb{Z}\}. \tag{1.1.4}$$

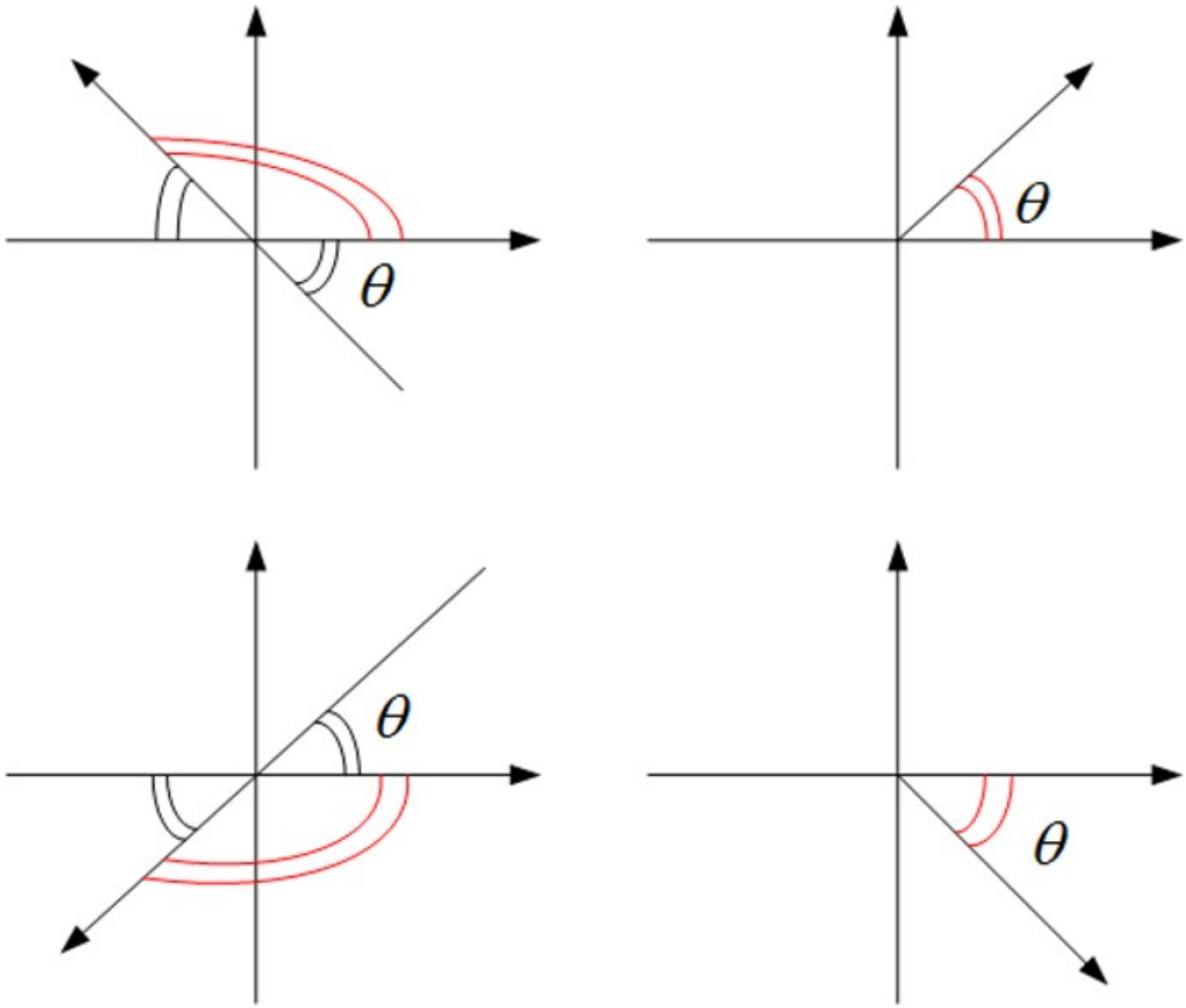
The argument of $z = 0$ is undefined.

For $z = x + iy \neq 0$, the principal argument can be determined from $\arctan(y/x)$ together with the quadrant (象限) of z:

$$\arg z (z \neq 0) = \begin{cases} \arctan\left(\dfrac{y}{x}\right) & (z \text{ is in the 1st or 4th quadrant}) \\ \arctan\left(\dfrac{y}{x}\right) + \pi & (z \text{ is in the 2nd quadrant}) \\ \arctan\left(\dfrac{y}{x}\right) - \pi & (z \text{ is in the 3rd quadrant}) \\ \dfrac{\pi}{2} & (z \text{ is on the positive imaginary axis}) \\ -\dfrac{\pi}{2} & (z \text{ is on the negative imaginary axis}) \\ \pi & (z \text{ is on the negative real axis}) \\ 0 & (z \text{ is on the positive real axis}) \end{cases} \tag{1.1.5}$$


y
1
−π
π
−3π/2
−π/2
O
π/2
3π/2
x
−1
y=tanx


y
π/2
y = arctan x
o
x
−π/2


θ
θ
θ
θ

Example 1.1

Find all arguments and the principal argument of $1+i$ *and* $-3\sqrt{3}-3i$.

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Solution.

For $z=1+i$, the point lies in the first quadrant, so its principal argument is

$$\arg z = \arctan\frac{y}{x} = \arctan 1 = \frac{\pi}{4}$$

The arguments are given by

$$\operatorname{Arg} z = \arg z + 2k\pi = \frac{\pi}{4} + 2k\pi \quad (k\in\mathbb{Z}).$$

Solution (Cont.)

For $z = -3\sqrt{3} - 3i$, the point lies in the third quadrant, so its principal argument is

$$\arg z = \arctan \frac{y}{x} - \pi = \arctan \frac{-3}{-3\sqrt{3}} - \pi = -\frac{5}{6}\pi.$$

All its arguments are

$$\operatorname{Arg} z = \arg z + 2k\pi = -\frac{5}{6}\pi + 2k\pi \quad (k \in \mathbb{Z}).$$

Using the relations between rectangular coordinates (直角坐标) and polar coordinates (极坐标), $x = r\cos\theta$ and $y = r\sin\theta$, we can write

$$z = r(\cos\theta + i\sin\theta) \tag{1.1.6}$$

where r is the modulus of z and θ is any argument of z. This is called the trigonometric form (三角形式) of z.

Using Euler's formula (欧拉公式) $e^{i\theta} = \cos\theta + i\sin\theta$, the trigonometric form becomes

$$z = re^{i\theta} \tag{1.1.7}$$

This is called the exponential form (指数形式) of z. Because θ may be replaced by $\theta + 2k\pi$, the trigonometric and exponential forms are not unique, even though they represent the same complex number.

Example 1.2

Express $z = 1 + \sin 1 + i \cos 1$ *in trigonometric and exponential forms.*

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Solution.

First compute the modulus r and the principal argument $\arg z$:

$$r^2 = (1+\sin 1)^2 + \cos^2 1 = 2(1+\sin 1)$$
$$= 2\left[1+\cos\left(\frac{\pi}{2}-1\right)\right] = 4\cos^2\left(\frac{\pi}{4}-\frac{1}{2}\right)$$
$$r = 2\cos\left(\frac{\pi}{4}-\frac{1}{2}\right) > 0$$

Solution (Cont.)

Since $1 + \sin 1 > 0$ and $\cos 1 > 0$, the point z lies in the first quadrant.

$$\begin{aligned}
\arg z &= \arctan \frac{y}{x} = \arctan \left(\frac{\cos 1}{1 + \sin 1} \right) \\
&= \arctan \left[\frac{\sin \left(\frac{\pi}{2} - 1 \right)}{1 + \cos \left(\frac{\pi}{2} - 1 \right)} \right] \\
&= \arctan \left[\frac{2 \sin \left(\frac{\pi}{4} - \frac{1}{2} \right) \cos \left(\frac{\pi}{4} - \frac{1}{2} \right)}{2 \cos^2 \left(\frac{\pi}{4} - \frac{1}{2} \right)} \right] \\
&= \left(\frac{\pi}{4} - \frac{1}{2} \right)
\end{aligned}$$

Solution (Cont.)

Therefore, the trigonometric form of z is

$$z = 2\cos\left(\frac{\pi}{4} - \frac{1}{2}\right)\left[\cos\left(\frac{\pi}{4} - \frac{1}{2}\right) + i\sin\left(\frac{\pi}{4} - \frac{1}{2}\right)\right]$$

Its exponential form is

$$z = 2\cos\left(\frac{\pi}{4} - \frac{1}{2}\right)e^{i\left(\frac{\pi}{4} - \frac{1}{2}\right)}$$

When writing a trigonometric or exponential form, it is usually convenient to choose $\theta = \arg z$.

On the nonuniqueness of arguments:

1. All arguments of $1+i$ are $\pi/4+2k\pi$, where $k \in \mathbb{Z}$;
2. Its exponential forms are $\sqrt{2}e^{i(\pi/4+2k\pi)}$, all of which have the same value;
3. Using the principal argument, we usually write the complex number simply as $\sqrt{2}e^{i\pi/4}$.

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1 Four Arithmetic Operations of Complex Numbers

The sum and difference of two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are defined by

$$(x_1 + iy_1) \pm (x_2 + iy_2) = (x_1 \pm x_2) + i(y_1 \pm y_2) \qquad (1.2.1)$$

Thus, we add or subtract the real parts and imaginary parts separately. When $y_1 = y_2 = 0$, these definitions reduce to the usual operations on real numbers.

Let z_1 and z_2 be represented by the vectors $\overrightarrow{OP_1}$ and $\overrightarrow{OP_2}$. Complex addition and subtraction agree with vector addition and subtraction. Hence, the diagonal (对角线) $\overrightarrow{OP}$ of the parallelogram (平行四边形) with adjacent sides (邻边) $\overrightarrow{OP_1}$ and $\overrightarrow{OP_2}$ represents $z_1 + z_2$, while $\overrightarrow{P_2P_1}$ represents $z_1 - z_2$ (see Figure 1.3).

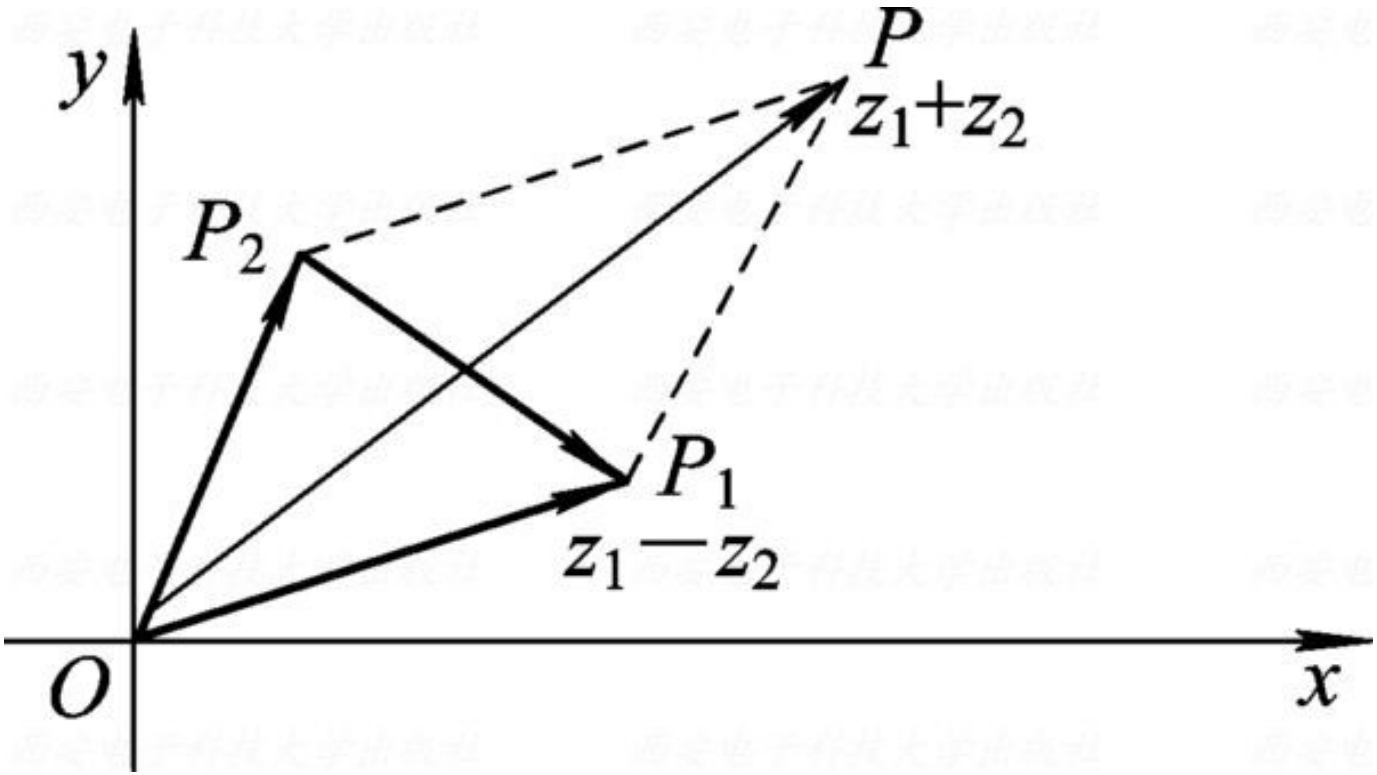


Figure: 1.3

By the parallelogram rule (平行四边形法则), the lengths $|z_1|$, $|z_2|$, and $|z_1 - z_2|$ form the side lengths of a triangle. In particular, $|z_1 - z_2|$ is the length of $\overrightarrow{P_2P_1}$, that is, the distance between the points z_1 and z_2. Indeed,

$$|z_1 - z_2| = |(x_1 - x_2) + i(y_1 - y_2)| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

which is precisely the distance formula in the plane.

The lengths $|z_1|$, $|z_2|$, and $|z_1 + z_2|$ also form the side lengths of a triangle. Therefore, the triangle inequality（三角不等式) and reverse triangle inequality（反三角不等式）give

$$|z_1 + z_2| \le |z_1| + |z_2|, \quad |z_1 - z_2| \ge ||z_1| - |z_2||$$

Multiplication and division are defined by

$$\begin{aligned} z_1 z_2 &= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1), \\ \frac{z_1}{z_2} &= \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2} \\ &= \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2} \quad (z_2 \neq 0). \end{aligned} \tag{1.2.2}$$

The multiplication and division formulas are especially transparent in trigonometric form. Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \\ &= r_1 r_2[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)] \\ &= r_1 r_2[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)] \end{aligned} \tag{1.2.3}$$

Similarly, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1(\cos\theta_1 + i\sin\theta_1)}{r_2(\cos\theta_2 + i\sin\theta_2)} = \frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)] \tag{1.2.4}$$

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, the corresponding exponential formulas are

$$\begin{aligned} z_1 z_2 &= r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1+\theta_2)} \\ \frac{z_1}{z_2} &= \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)} \quad (r_2 \neq 0) \end{aligned} \tag{1.2.5}$$

Consequently,

$$|z_1 z_2| = |z_1|\,|z_2|\,, \quad \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|} \tag{1.2.6}$$

For principal arguments, the corresponding relations are understood modulo 2π:

$$\begin{cases} \arg(z_1 z_2) \equiv \arg z_1 + \arg z_2 \pmod{2\pi}, \\ \arg\left(\dfrac{z_1}{z_2}\right) \equiv \arg z_1 - \arg z_2 \pmod{2\pi}. \end{cases} \tag{1.2.7}$$

For example, if $z_1 = -1$ and $z_2 = i$, then $z_1 z_2 = -i$, so

$$\operatorname{Arg} z_1 = \pi + 2n\pi \quad (n \in \mathbb{Z})$$

$$\operatorname{Arg} z_2 = \frac{\pi}{2} + 2m\pi \quad (m \in \mathbb{Z})$$

$$\operatorname{Arg} z_1 z_2 = -\frac{\pi}{2} + 2k\pi \quad (k \in \mathbb{Z})$$

Choosing arbitrary representatives of these arguments, we obtain

$$\frac{3\pi}{2} + 2(m+n)\pi = -\frac{\pi}{2} + 2k\pi$$

This equality holds exactly when $k = m+n+1$. Thus, for any chosen $m, n \in \mathbb{Z}$, an appropriate k always exists, and conversely. For example, if $m = n = 0$, then $k = 1$; if $k = -1$, one may take $(m, n) = (0, -2)$ or $(-2, 0)$.

The trigonometric and exponential forms reveal the geometric meaning of multiplication. The vector representing $z_1 z_2$ is obtained from the vector representing z_1 by scaling (尺度变换) its length by $r_2 = |z_2|$ and rotating (旋转) it through an argument of z_2 (see Figure 1.4).

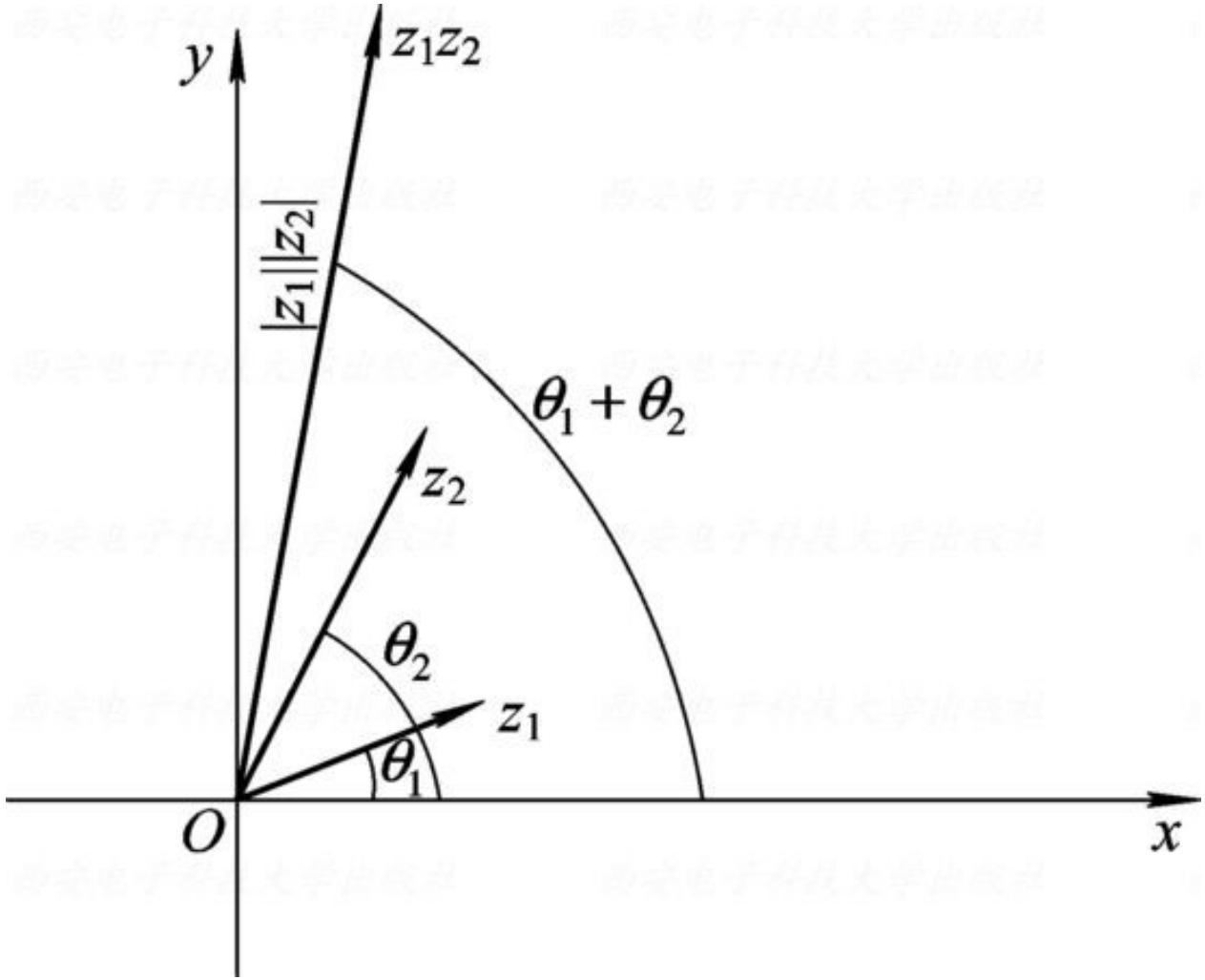


Figure: 1.4

- Which expression scales the modulus of z by $\sqrt{2}$ and rotates its vector counterclockwise through $\frac{\pi}{4}$?

$$\text{A.}z\sqrt{2}e^{-i\frac{\pi}{4}} \qquad \text{B.}\frac{z}{\sqrt{2}}e^{-i\frac{\pi}{4}} \qquad \text{C.}\frac{z}{\sqrt{2}}e^{i\frac{\pi}{4}} \qquad \text{D.}z\sqrt{2}e^{i\frac{\pi}{4}}$$

- The expression $\dfrac{z}{1-\sqrt{3}i}$ scales the modulus of z by _____ and changes its argument by _____.
- Let $z_3 = \dfrac{z_1}{z_2}$. Suppose $|z_3| = \dfrac{1}{5}|z_1|$ and the argument of z_3 is obtained by rotating the argument of z_1 counterclockwise through $\dfrac{\pi}{6}$. Find z_2.

Example 2.1

Let z_1 and z_2 be arbitrary points in the complex plane. Prove that $|z_1 - z_2| \geq ||z_1| - |z_2||$.

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Proof.

Geometrically, the inequality states that one side of a triangle is at least the absolute difference of the other two side lengths.
Algebraically, it follows from the triangle inequality
$|z_1 + z_2| \leq |z_1| + |z_2|$.
Because

$$|z_1| = |z_1 - z_2 + z_2| \leq |z_1 - z_2| + |z_2|$$

Proof (Cont.)

$$|z_1| - |z_2| \leq |z_1 - z_2| \tag{1}$$

Because

$$|z_2| = |z_2 - z_1 + z_1| \leq |z_2 - z_1| + |z_1|$$

$$|z_2| - |z_1| \leq |z_2 - z_1| \tag{2}$$

Using inequalities (1) and (2), we have

$$|z_1 - z_2| \geq ||z_1| - |z_2||$$



Example 2.2

Two vertices（顶点） of an equilateral triangle（等边三角形） are $z_1 = 1 + 2i$ *and* $z_2 = 3 + 2i$. *Find the possible third vertices.*

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Solution.

As shown in Figure 1.5, fix z_1 and rotate the vector $z_2 - z_1$ through $\frac{\pi}{3}$ or $-\frac{\pi}{3}$. The endpoint (终点) of the resulting vector is a possible third vertex z_3:

$$z_3 = z_1 + (z_2 - z_1)e^{\pm i\pi/3} = (1+2i) + 2\left(\frac{1}{2} \pm i\frac{\sqrt{3}}{2}\right) = 2 + (2 \pm \sqrt{3})i.$$

Solution (Cont.)

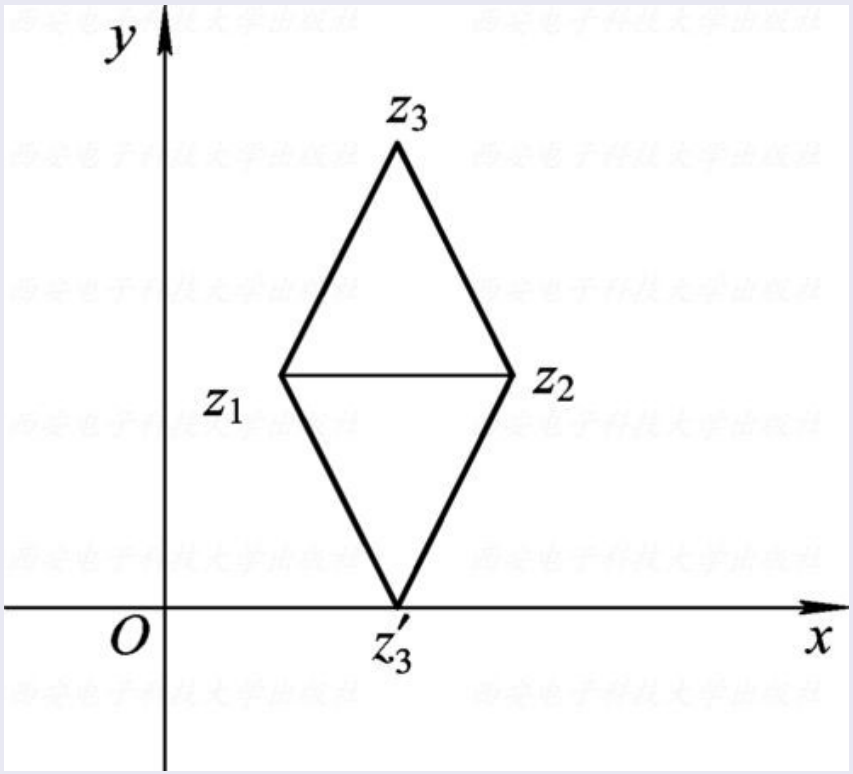


Figure: 1.5

As with real numbers, complex addition and multiplication satisfy the commutative, associative, and distributive laws:

$$z_1 + z_2 = z_2 + z_1;\ \ z_1 z_2 = z_2 z_1 \qquad \text{(commutative law)}$$

$$z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3;\ \ z_1(z_2 z_3) = (z_1 z_2) z_3 \qquad \text{(associative law)}$$

$$z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3 \qquad \text{(distributive law)}$$

Together with these operations, the set of complex numbers forms the complex field（复数域）, denoted by $\mathbb{C}$.

2 Powers and Roots of Complex Numbers

For $n \in \mathbb{N}$, the product of n copies of z is called the n-th power of z and is denoted by z^n:

$$z^n = \underbrace{z \cdot z \cdots z}_{n \text{ factors}}.$$

If $z = re^{i\theta}$, then

$$z^n = r^n e^{in\theta} = r^n(\cos n\theta + i \sin n\theta) \quad (n \in \mathbb{N})$$

In particular, when $r = 1$, de Moivre's formula (棣莫弗公式) is obtained:

$$(\cos\theta + i\sin\theta)^n = (\cos n\theta + i\sin n\theta) \tag{1.2.8}$$

For $n \in \mathbb{N}$, we have $z^n = r^n(\cos n\theta + i \sin n\theta)$. If $z \neq 0$, negative integer powers are defined by $z^{-n} = 1/z^n$. Thus,

$$\begin{aligned} z^{-n} &= \frac{1}{r^n e^{in\theta}} = r^{-n} e^{-in\theta} \\ &= r^{-n}[\cos(-n\theta) + i \sin(-n\theta)]. \end{aligned}$$

Therefore, $z^m = r^m(\cos m\theta + i \sin m\theta)$ for every $m \in \mathbb{Z}$, provided $z \neq 0$ when $m < 0$.

Let $z \neq 0$. If $\omega^n = z$, then ω is called an n-th root (n 次方根) of z. Write $z = re^{i\theta} = r(\cos\theta + i\sin\theta)$ and $\omega = \rho e^{i\varphi} = \rho(\cos\varphi + i\sin\varphi)$. Using the power formula for the trigonometric form together with $\omega^n = z$, we obtain

$$\rho^n(\cos n\varphi + i\sin n\varphi) = r(\cos\theta + i\sin\theta) \tag{1.2.9}$$

Thus, $\rho^n = r$, $\cos n\varphi = \cos\theta$, and $\sin n\varphi = \sin\theta$. The necessary and sufficient conditions for (1.2.9) are

$$\rho^n = r, \quad n\varphi = \theta + 2k\pi \quad (k \in \mathbb{Z})$$

Therefore, $\rho = r^{1/n}$ and $\varphi = (\theta + 2k\pi)/n$, where $r^{1/n}$ is the positive real n-th root of r. Hence,

$$\omega_k = r^{1/n}\left(\cos\frac{\theta + 2k\pi}{n} + i\sin\frac{\theta + 2k\pi}{n}\right). \tag{1.2.10}$$

For $k = 0, 1, 2, \ldots, n-1$, formula (1.2.10) gives n distinct roots:

$$\omega_0 = r^{\frac{1}{n}} \left(\cos \frac{\theta}{n} + i \sin \frac{\theta}{n} \right)$$

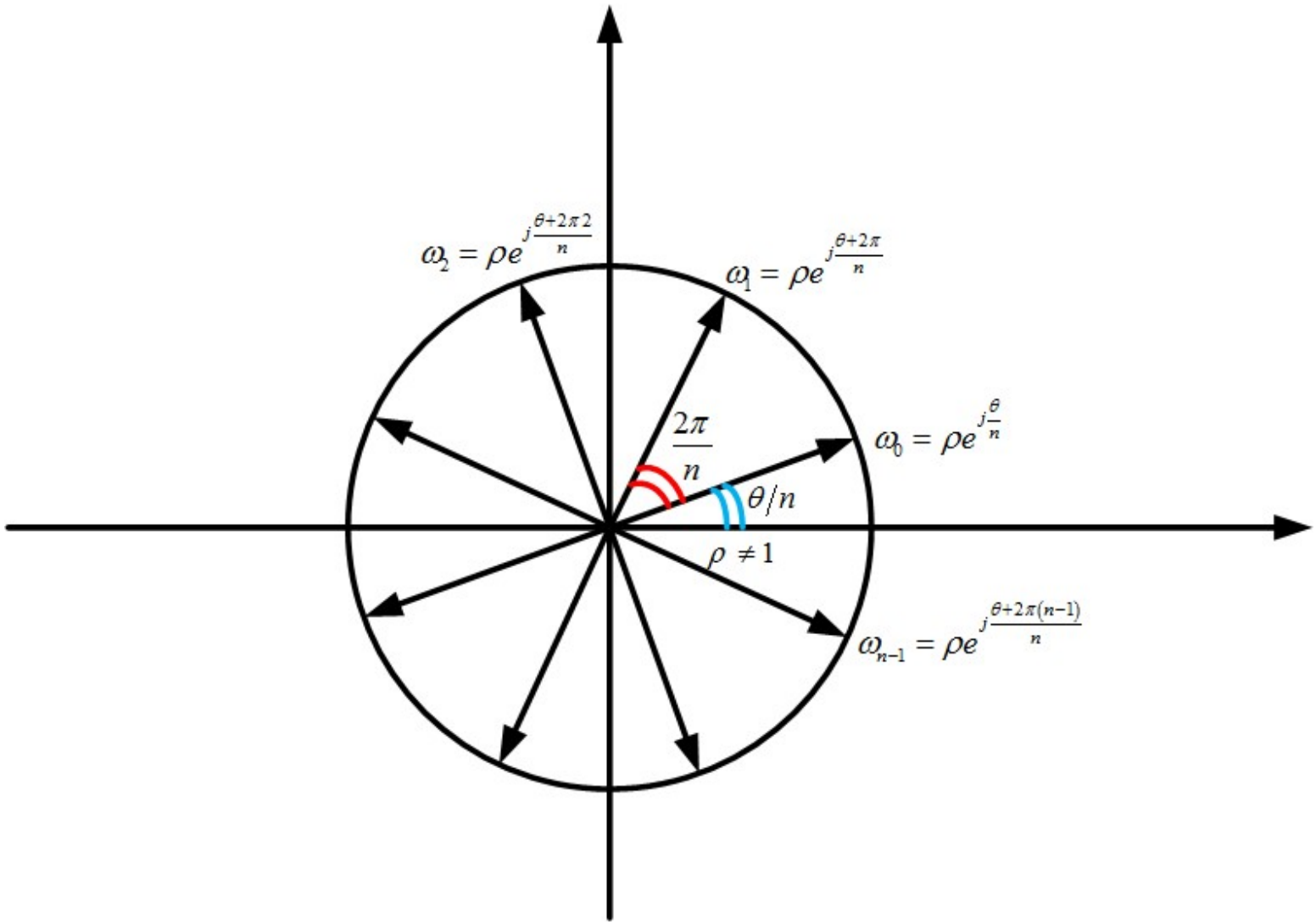
$$\omega_1 = r^{\frac{1}{n}} \left(\cos \frac{\theta + 2\pi}{n} + i \sin \frac{\theta + 2\pi}{n} \right)$$

$$\vdots$$

$$\omega_{n-1} = r^{\frac{1}{n}} \left(\cos \frac{\theta + 2(n-1)\pi}{n} + i \sin \frac{\theta + 2(n-1)\pi}{n} \right)$$

Other integer values of k repeat these roots. For example,

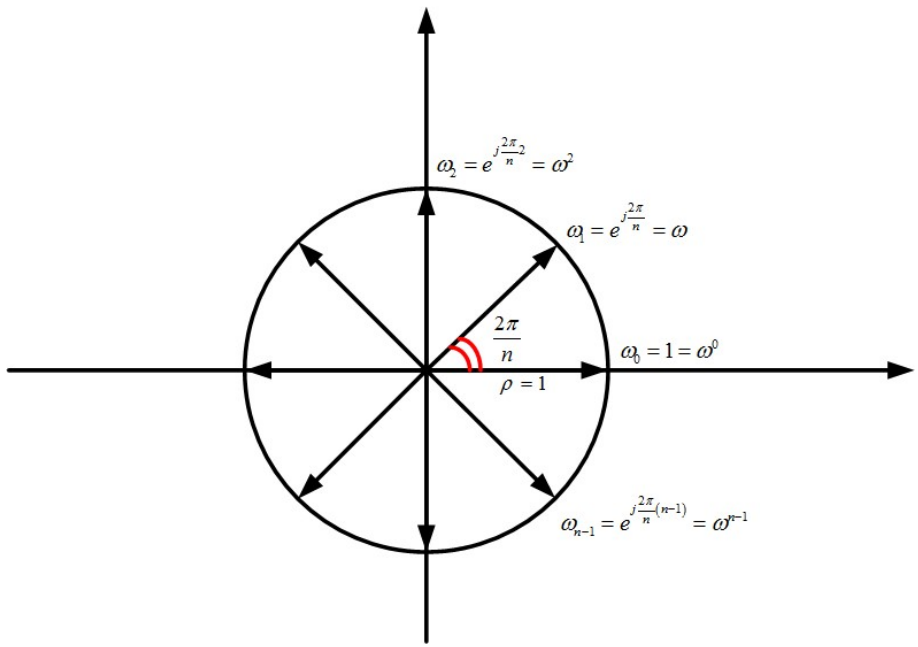
$$\omega_n = r^{\frac{1}{n}} \left(\cos \frac{\theta + 2n\pi}{n} + i \sin \frac{\theta + 2n\pi}{n} \right) = \omega_0$$


$\omega_2 = \rho e^{j\frac{\theta+2\pi 2}{n}}$
$\omega_1 = \rho e^{j\frac{\theta+2\pi}{n}}$
$\omega_0 = \rho e^{j\frac{\theta}{n}}$
$\frac{2\pi}{n}$
θ/n
$\rho \neq 1$
$\omega_{n-1} = \rho e^{j\frac{\theta+2\pi(n-1)}{n}}$

In particular, when $z = 1$, let

$$\omega = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$$

Then the n-th roots of unity（单位根） are $1, \omega, \omega^2, \ldots, \omega^{n-1}$.



Example 2.3

Evaluate $\left(\dfrac{\sqrt{3}+i}{1-i}\right)^6$.

Example 2.3

Evaluate $\left(\dfrac{\sqrt{3}+i}{1-i}\right)^6$.

Solution.

Because

$$\sqrt{3}+i = 2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right) = 2e^{\frac{\pi}{6}i}$$

$$1-i = \sqrt{2}\left[\cos\left(-\frac{\pi}{4}\right)+i\sin\left(-\frac{\pi}{4}\right)\right] = \sqrt{2}e^{-\frac{\pi}{4}i}$$

Solution (Cont.)

$$\frac{\sqrt{3}+i}{1-i} = \frac{2e^{\frac{\pi}{6}i}}{\sqrt{2}e^{-\frac{\pi}{4}i}} = \sqrt{2}e^{\frac{5\pi}{12}i}$$

Therefore,

$$\left(\frac{\sqrt{3}+i}{1-i}\right)^6 = \left(\sqrt{2}e^{\frac{5\pi}{12}i}\right)^6 = 8e^{\frac{5\pi}{2}i} = 8\left(\cos\frac{5\pi}{2} + i\sin\frac{5\pi}{2}\right) = 8i$$

Example 2.4

Solve the equation $(1+z)^6 = (1-z)^6$.

Example 2.4

Solve the equation $(1+z)^6 = (1-z)^6$.

Solution.

Since $z = 1$ is not a solution, we may divide by $(1-z)^6$ and write

$$\left(\frac{1+z}{1-z}\right)^6 = 1$$

Let $\omega = \dfrac{1+z}{1-z}$, then $\omega^6 = 1$.

Solution (Cont.)

Since $1 = \cos 0 + i \sin 0$, the sixth roots of unity are

$$\sqrt[6]{1} = \cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6} \quad (k = 0, 1, 2, 3, 4, 5)$$

then

$$\omega_0 = (\cos 0 + i \sin 0)$$

$$\omega_1 = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$$

Solution (Cont.)

$$\omega_2 = \left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

$$\omega_3 = (\cos\pi + i\sin\pi)$$

$$\omega_4 = \left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$$

$$\omega_5 = \left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right)$$

These six roots are the vertices of a regular hexagon（正六边形) inscribed in（内接于）the unit circle (see Figure 1.6).

Solution (Cont.)

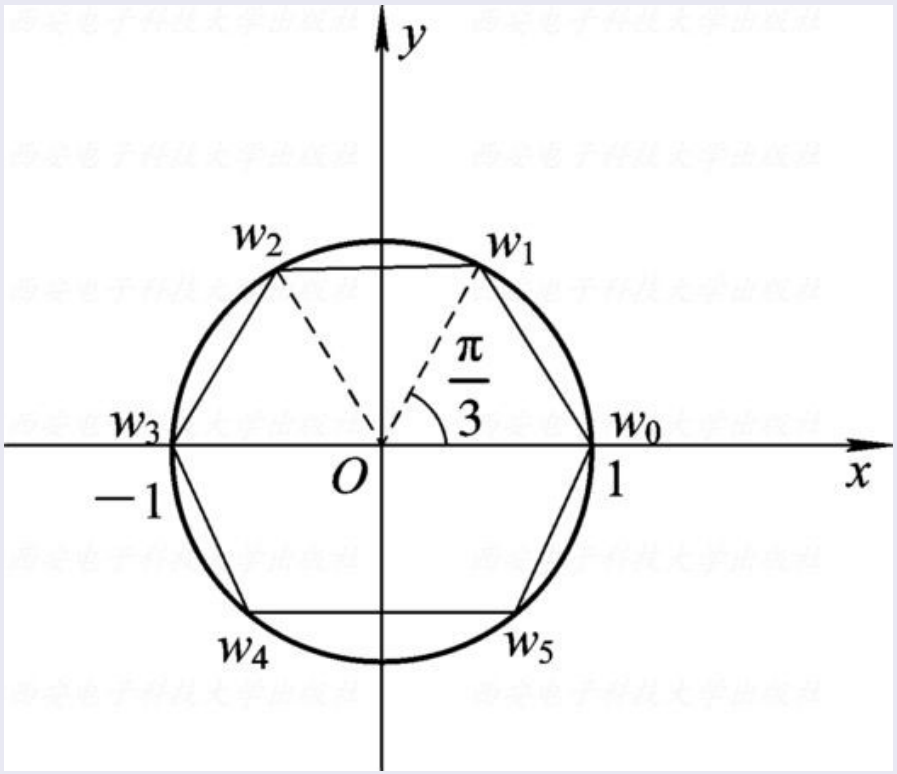


Figure: 1.6

Solution (Cont.)

Write the sixth roots as $\omega = e^{i\alpha}$, where $\alpha = \dfrac{k\pi}{3}$ $(k = 0, 1, \ldots, 5)$.

From

$$\omega = \frac{1+z}{1-z} \Rightarrow z = \frac{\omega - 1}{\omega + 1}$$

$$z = \frac{e^{i\alpha} - 1}{e^{i\alpha} + 1} = \frac{\cos\alpha + i\sin\alpha - 1}{\cos\alpha + i\sin\alpha + 1}$$

$$z = \frac{2\sin\frac{\alpha}{2}\left(-\sin\frac{\alpha}{2} + i\cos\frac{\alpha}{2}\right)}{2\cos\frac{\alpha}{2}\left(\cos\frac{\alpha}{2} + i\sin\frac{\alpha}{2}\right)} = \tan\frac{\alpha}{2}\frac{e^{i\left(\frac{\pi}{2}+\frac{\alpha}{2}\right)}}{e^{i\frac{\alpha}{2}}} = i\tan\frac{\alpha}{2}$$

The value $\alpha = \pi$ corresponds to $\omega = -1$, for which $\omega + 1 = 0$, so it yields no finite z. Therefore, the solutions are

$$z = i\tan\frac{\alpha}{2}, \qquad \alpha = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}.$$

3 Complex Conjugate

For $z = x + iy$, its complex conjugate (共轭复数) is $\bar{z} = x - iy$. Geometrically, the points z and $\bar{z}$ are symmetric (对称) about the real axis (see Figure 1.7). Hence, $|z| = |\bar{z}|$. If $z \neq 0$ and z is not on the negative real axis, then $\arg \bar{z} = -\arg z$.

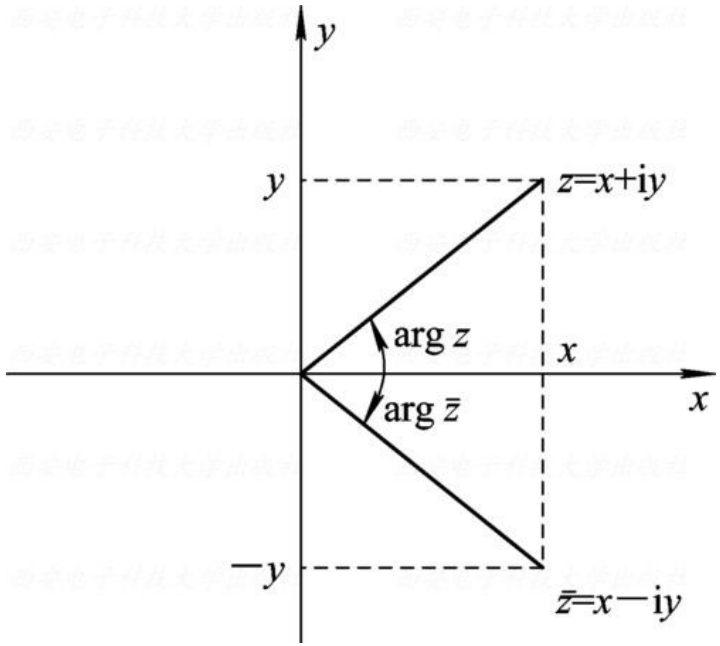


Figure: 1.7

Some useful properties of complex conjugation are

1. $|z| = |\bar{z}|$, $\arg \bar{z} \equiv -\arg z \pmod{2\pi}$ for $z \neq 0$, and $\bar{\bar{z}} = z$;
2. $z\bar{z} = [\mathrm{Re}(z)]^2 + [\mathrm{Im}(z)]^2 = |z|^2$
3. $z + \bar{z} = 2\,\mathrm{Re}(z)$ and $z - \bar{z} = 2i\,\mathrm{Im}(z)$;
4. $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$, $\overline{z_1 z_2} = \bar{z}_1\,\bar{z}_2$, and $\overline{(z_1/z_2)} = \bar{z}_1/\bar{z}_2$ for $z_2 \neq 0$;
5. If P and Q are polynomials with real coefficients and $Q(z) \neq 0$, then $\overline{\left(\dfrac{P(z)}{Q(z)}\right)} = \dfrac{\overline{P(z)}}{\overline{Q(z)}} = \dfrac{P(\bar{z})}{Q(\bar{z})}$.

Example 2.5

Let $z_1 = 1 + 2i$ *and* $z_2 = -3 - 4i$. *Find* $\dfrac{z_1}{z_2}$, $\overline{\left(\dfrac{z_1}{z_2}\right)}$, $\mathrm{Re}\left(\dfrac{z_1}{z_2}\right)$, *and* $\mathrm{Im}\left(\dfrac{z_1}{z_2}\right)$.

Example 2.5

Let $z_1 = 1 + 2i$ *and* $z_2 = -3 - 4i$. *Find* $\dfrac{z_1}{z_2}$, $\overline{\left(\dfrac{z_1}{z_2}\right)}$, $\mathrm{Re}\left(\dfrac{z_1}{z_2}\right)$, *and* $\mathrm{Im}\left(\dfrac{z_1}{z_2}\right)$.

Solution.

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{1+2i}{-3-4i} = \frac{(1+2i)(-3+4i)}{(-3-4i)(-3+4i)} \\ &= \frac{-3-8+(4-6)i}{25} = -\frac{11}{25} - \frac{2}{25}i\end{aligned}$$

Solution (Cont.)

Therefore,

$$\overline{\left(\frac{z_1}{z_2}\right)} = -\frac{11}{25} + \frac{2}{25}i$$

and hence

$$\mathrm{Re}\left(\frac{z_1}{z_2}\right) = -\frac{11}{25}, \quad \mathrm{Im}\left(\frac{z_1}{z_2}\right) = -\frac{2}{25}$$

Example 2.6

Let $a, b \in \mathbb{R}$, *with* $b \neq 0$ *and* $|a + ib| = 1$. *Prove that there exists* $c \in \mathbb{R}$ *such that* $a + ib = \dfrac{c + i}{c - i}$.

Proof.

$$
\begin{aligned}
& z = a + ib \\
\Rightarrow\ & z = \frac{c+i}{c-i} \\
\Rightarrow\ & c = \frac{z+1}{z-1} i \\
\Rightarrow\ & c = \frac{a+ib+1}{a+ib-1} i = \frac{(a+1)+ib}{(a-1)+ib} i = \frac{[(a+1)+ib][(a-1)-ib]}{[(a-1)+ib][(a-1)-ib]} i \\
& = \frac{(a+1)(a-1)+b^2+ib[(a-1)-(a+1)]}{(a-1)^2+b^2} i \\
& = \frac{a^2+b^2-1-2bi}{a^2-2a+1+b^2} i \\
& = \frac{b}{(1-a)} \qquad (\because a^2+b^2=1)
\end{aligned}
$$

□

Example 2.7

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be arbitrary complex numbers. Prove that

1. $z_1\overline{z_2} + \overline{z_1}z_2 = 2\,\mathrm{Re}(z_1\overline{z_2})$;
2. $|z_1 + z_2| \leq |z_1| + |z_2|$.

Proof.

1 $$\begin{aligned} z_1\overline{z_2} + \overline{z_1}z_2 &= (x_1 + iy_1)(x_2 - iy_2) + (x_1 - iy_1)(x_2 + iy_2) \\ &= (x_1x_2 + y_1y_2) + i(x_2y_1 - x_1y_2) \\ &\quad + (x_1x_2 + y_1y_2) + i(-x_2y_1 + x_1y_2) \\ &= 2(x_1x_2 + y_1y_2) \\ &= 2\,\mathrm{Re}(z_1\overline{z_2}) \end{aligned}$$

or $z_1\overline{z_2} + \overline{z_1}z_2 = z_1\overline{z_2} + \overline{z_1\overline{z_2}} = 2\,\mathrm{Re}(z_1\overline{z_2})$

Proof (Cont.)

2. Although the triangle inequality has a geometric interpretation, we now prove it algebraically. Since

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2)(\overline{z_1 + z_2}) \\ &= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) \\ &= z_1\overline{z_1} + z_2\overline{z_2} + \overline{z_1}z_2 + z_1\overline{z_2} \\ &= |z_1|^2 + |z_2|^2 + \overline{z_1}z_2 + z_1\overline{z_2} \end{aligned}$$

Proof (Cont.)

Using $z_1\overline{z_2} + \overline{z_1}z_2 = 2\,\mathrm{Re}(z_1\overline{z_2})$, we obtain

$$\begin{aligned} |z_1 + z_2|^2 &= |z_1|^2 + |z_2|^2 + 2\,\mathrm{Re}(z_1\overline{z_2}) \\ &\leq |z_1|^2 + |z_2|^2 + 2\,|z_1\overline{z_2}| \\ &= |z_1|^2 + |z_2|^2 + 2\,|z_1|\,|z_2| \\ &= (|z_1| + |z_2|)^2 \end{aligned}$$

Taking square roots on both sides completes the proof of the triangle inequality. □

Example 2.8

Prove that three pairwise distinct complex numbers z_1, z_2, and z_3 are the vertices of an equilateral triangle if and only if

$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1.$$

Proof.

Method 1.
The triangle $\triangle z_1 z_2 z_3$ is equilateral if and only if rotating $\overrightarrow{z_1 z_2}$ through $\pm\frac{\pi}{3}$ about z_1 gives $\overrightarrow{z_1 z_3}$. Thus, $z_3 - z_1 = (z_2 - z_1)e^{\pm i\pi/3}$, or

Proof (Cont.)

$$\frac{z_3 - z_1}{z_2 - z_1} = e^{\pm i\pi/3} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2}, \quad \frac{z_3 - z_1}{z_2 - z_1} - \frac{1}{2} = \pm i\frac{\sqrt{3}}{2}.$$

Squaring both sides and simplifying gives

$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$$



Proof.

Method 2.

The equation $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$ is equivalent to

$$\frac{z_2 - z_1}{z_3 - z_1} = \frac{z_1 - z_3}{z_2 - z_3} \tag{1}$$

If z_1, z_2, and z_3 satisfy (1), then taking absolute values after cross-multiplication gives

$$|z_3 - z_1|^2 = |z_1 - z_2|\,|z_2 - z_3| \tag{2}$$

Proof (Cont.)

$$\frac{z_2 - z_1}{z_3 - z_1} - 1 = \frac{z_1 - z_3}{z_2 - z_3} - 1$$

which is equivalent to

$$\frac{z_2 - z_3}{z_3 - z_1} = \frac{z_1 - z_2}{z_2 - z_3} \tag{3}$$

Taking absolute values gives

$$|z_2 - z_3|^2 = |z_1 - z_2|\,|z_3 - z_1| \tag{4}$$

Proof (Cont.)

Let $a = |z_2 - z_3|$, $b = |z_3 - z_1|$, and $c = |z_1 - z_2|$. These are positive because the three points are pairwise distinct. Then (2) and (4) read

$$b^2 = ca, \qquad a^2 = cb.$$

Dividing the first equation by the second gives $\dfrac{b^2}{a^2} = \dfrac{a}{b}$, so $a^3 = b^3$ and hence $a = b$. Substituting into $a^2 = cb$ gives $c = a$. Therefore,

$$|z_2 - z_3| = |z_3 - z_1| = |z_1 - z_2|,$$

that is, $\triangle z_1 z_2 z_3$ is equilateral.

Proof (Cont.)

Conversely, if $\triangle z_1 z_2 z_3$ is equilateral, then

$$\left|\frac{z_2 - z_1}{z_3 - z_1}\right| = \left|\frac{z_1 - z_3}{z_2 - z_3}\right| = 1.$$

Moreover, every interior angle equals $\dfrac{\pi}{3}$, so

$$\frac{z_2 - z_1}{z_3 - z_1} = e^{i\sigma_1 \pi/3}, \qquad \frac{z_1 - z_3}{z_2 - z_3} = e^{i\sigma_2 \pi/3}, \qquad \sigma_1, \sigma_2 \in \{1, -1\},$$

where, by (1.2.7),

$$\sigma_1 \frac{\pi}{3} \equiv \arg(z_2 - z_1) - \arg(z_3 - z_1) \pmod{2\pi},$$
$$\sigma_2 \frac{\pi}{3} \equiv \arg(z_1 - z_3) - \arg(z_2 - z_3) \pmod{2\pi}.$$

Proof (Cont.)

Traverse the triangle along the cycle $z_1 \to z_3 \to z_2 \to z_1$. The first quotient is the rotation at z_1 from the side toward the next vertex z_3 to the side toward the previous vertex z_2; the second quotient is the same rotation at z_3, from the side toward z_2 to the side toward z_1. Hence both rotations have the same orientation: $\sigma_1 = \sigma_2 = 1$ if the cycle is counterclockwise, and $\sigma_1 = \sigma_2 = -1$ otherwise. Therefore,

$$\frac{z_2 - z_1}{z_3 - z_1} = \frac{z_1 - z_3}{z_2 - z_3}$$

□

Finally, we illustrate two geometric applications of complex numbers: (i) expressing a plane curve (平面曲线) $F(x, y) = 0$ by a complex equation, and (ii) identifying the curve or region represented by a complex equation.

Example 2.9

Find a complex equation for the straight line（直线） through $z_1 = x_1 + iy_1$ *and* $z_2 = x_2 + iy_2$.

Example 2.9

Find a complex equation for the straight line（直线）*through* $z_1 = x_1 + iy_1$ *and* $z_2 = x_2 + iy_2$.

Solution.

The line through the distinct points (x_1, y_1) and (x_2, y_2) has the following system of parametric equations（参数方程组）:

$$\begin{cases} x = x_1 + t(x_2 - x_1) \\ y = y_1 + t(y_2 - y_1) \end{cases} \qquad (t \in \mathbb{R}).$$

Therefore, with $z = x + iy$, the corresponding complex parametric equation（复参数方程）is

$$z = (x_1 + iy_1) + t[(x_2 + iy_2) - (x_1 + iy_1)] = z_1 + t(z_2 - z_1).$$

Solution (Cont.)

Alternatively, write the line in general form (一般式) as $ax + by = c$, where $a, b, c \in \mathbb{R}$ and $(a, b) \neq (0, 0)$. If $z = x + iy$, then

$$x = \frac{z + \bar{z}}{2}, \quad y = \frac{z - \bar{z}}{2i}$$

Substitution gives

$$a\left(\frac{z + \bar{z}}{2}\right) + b\left(\frac{z - \bar{z}}{2i}\right) = c$$

which can be simplified as

$$(a - ib)z + (a + ib)\bar{z} = 2c$$

Example 2.10

Describe the loci（轨迹）or point sets（点集）represented by the following conditions:

1. $|z+2i| = 2$;
2. $|z-i| + |z+i| \leq 4$;
3. $0 \leq \arg(z-1) \leq \dfrac{\pi}{4}$ *and* $2 \leq \operatorname{Re} z \leq 3$.

Example 2.10

Describe the loci (轨迹) *or point sets* (点集) *represented by the following conditions:*

1. $|z+2i| = 2$;
2. $|z-i| + |z+i| \leq 4$;
3. $0 \leq \arg(z-1) \leq \dfrac{\pi}{4}$ *and* $2 \leq \operatorname{Re} z \leq 3$.

Solution.

1. The equation $|z+2i| = 2$ describes all points at distance 2 from $-2i$. Hence, it represents the circle (圆) centered at $-2i$ with radius (半径) 2 (see Figure 1.8(a)).

Solution (Cont.)

2. The sum of the distances from z to the foci (焦点) i and $-i$ is at most 4. This is an ellipse with $2a = 4$ and focal distance $2c = 2$, so $b^2 = a^2 - c^2 = 3$. Hence, the condition represents the closed elliptical region

$$\frac{x^2}{3} + \frac{y^2}{4} \leq 1$$

including its boundary (see Figure 1.8(b)).

3. The condition $0 \leq \arg(z-1) \leq \dfrac{\pi}{4}$ describes the closed sector with vertex 1 bounded by the rays of angles 0 and $\dfrac{\pi}{4}$, that is, $0 \leq y \leq x - 1$. Intersecting it with the strip $2 \leq x \leq 3$ gives the closed trapezoid (梯形) with vertices 2, 3, $3+2i$, and $2+i$ (see Figure 1.8(c)).

Solution (Cont.)

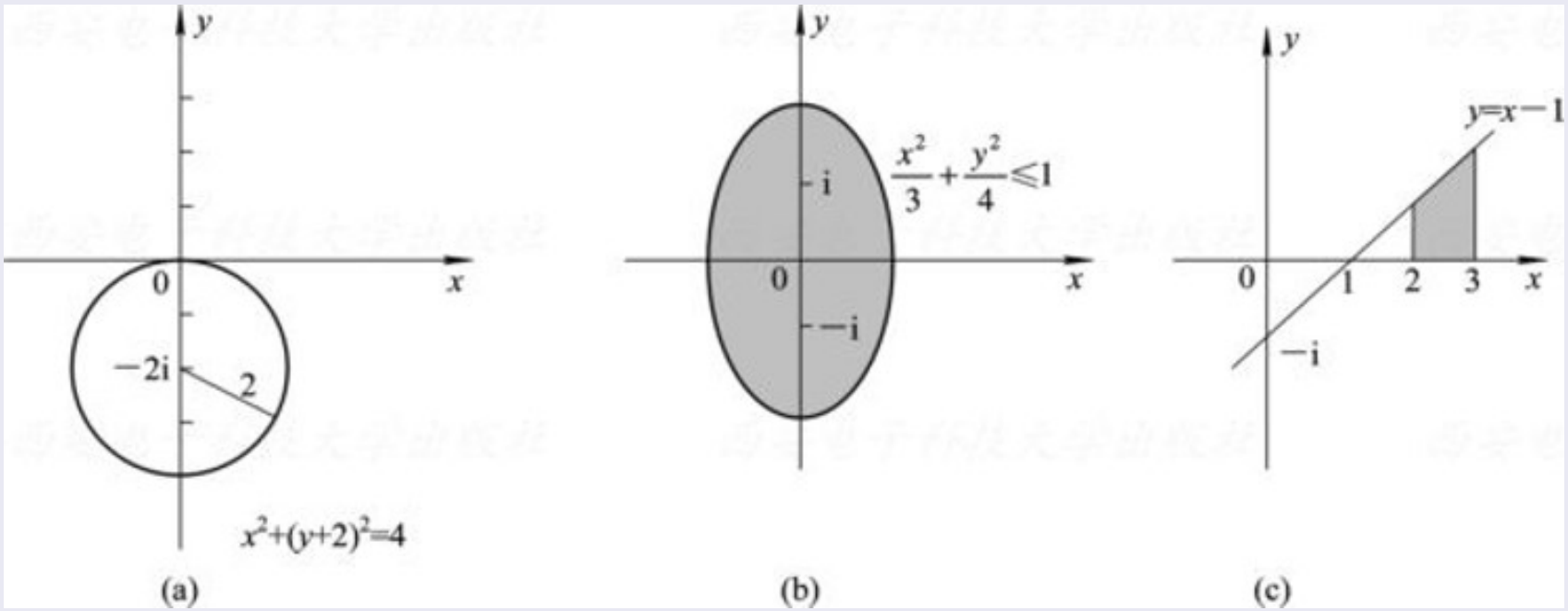


Figure: 1.8

4 The Riemann Sphere and the Point at Infinity

Consider a sphere (球面) tangent to (相切于) the complex plane at the origin O. Let the point of tangency (切点) be the south pole (南极) S, and let the perpendicular diameter (垂直直径) through S meet the sphere again at the north pole (北极) N (see Figure 1.9). For each point z in the complex plane, the line through N and z meets the sphere at exactly one additional point P. Conversely, every point $P \neq N$ determines exactly one point z in the plane. This correspondence is called stereographic projection (球极投影), and the sphere is called the Riemann sphere (复球面).

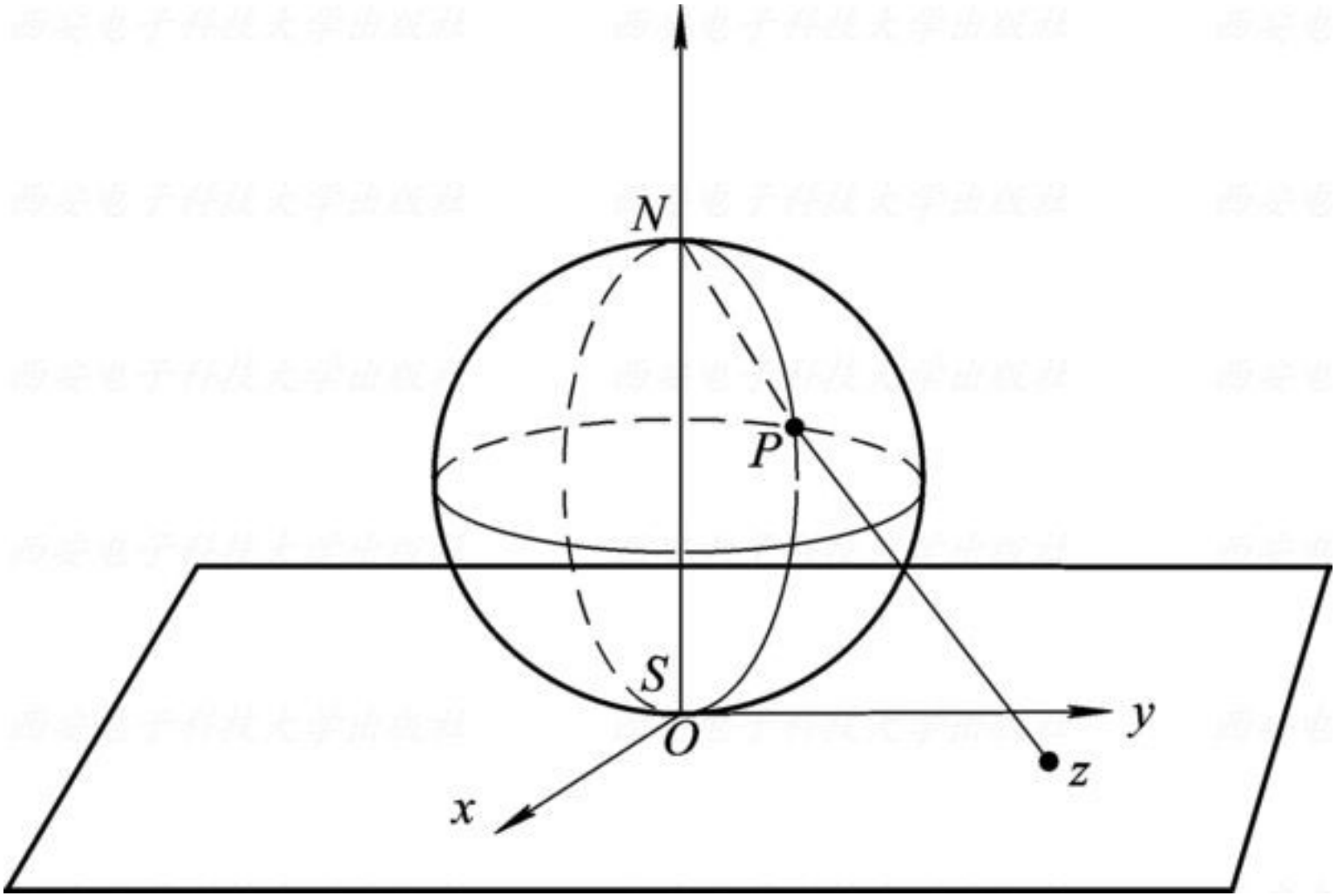


Figure: 1.9

The complex plane together with the point at infinity (无穷远点) ∞ is called the extended complex plane (扩充复平面). Under stereographic projection, finite points correspond to points on the sphere other than N, while ∞ corresponds to N. Thus, the Riemann sphere is a geometric model of the extended complex plane.

There is only one point at infinity in the extended complex plane; it is not separated into $+\infty$ and $-\infty$ as on the extended real line. The point ∞ is not an ordinary complex number, so its real part, imaginary part, and argument are undefined. With the extended-modulus convention, one writes $|\infty| = +\infty$.

For a finite complex number a, arithmetic involving ∞ is only partially defined:

- Addition and subtraction: $a + \infty = \infty + a = \infty$ and $a - \infty = \infty - a = \infty$;
- Multiplication: $a \cdot \infty = \infty \cdot a = \infty$ for $a \neq 0$;
- Division: $a/\infty = 0$ for finite a, while $\infty/a = \infty$ and $a/0 = \infty$ for finite $a \neq 0$;
- The forms $\infty - \infty$, $0 \cdot \infty$, ∞/∞, and $0/0$ are undefined.

Outline

1. Complex Numbers and Their Expressions
2. Operations of Complex Numbers
3. Domains in the Complex Plane
4. Complex Functions
5. Limits and Continuity of Complex Functions

1 Concepts of Point Sets in the Complex Plane

We begin with a point z_0 in the complex plane.

Definition

For $\delta > 0$, the open disk centered at z_0 with radius δ is

$$N_\delta(z_0) = \{z : |z - z_0| < \delta\}.$$

It is called the δ-neighborhood (δ 邻域) of z_0 (see Figure 1.10). The set

$$N_\delta^*(z_0) = \{z : 0 < |z - z_0| < \delta\}$$

is called the deleted δ-neighborhood (δ 去心邻域) of z_0.

Definition (Cont.)

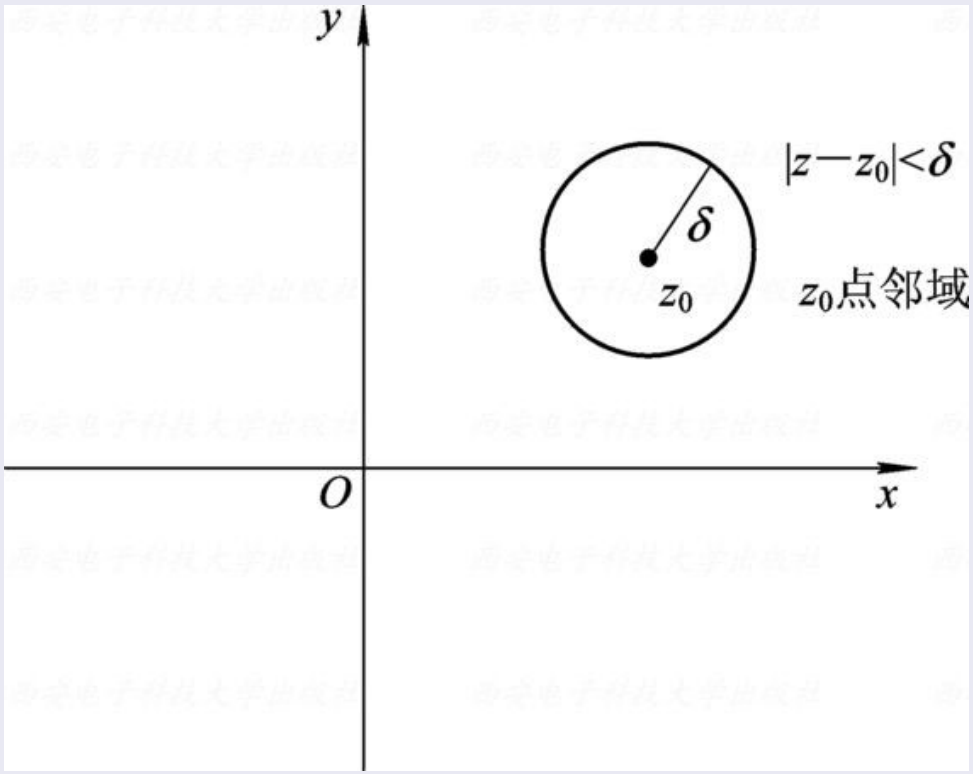


Figure: *1.10*

Definition

Let G be a point set (点集) in the complex plane.

- *A point $z_0 \in G$ is an interior point (内点) of G if some neighborhood of z_0 is contained in G.*
- *The set G is an open set (开集) if every point of G is an interior point.*
- *A point $z_0 \in \mathbb{C}$ is a boundary point (边界点) of G if every neighborhood of z_0 contains both points of G and points not in G.*
- *The set of all boundary points is the boundary (边界) of G, denoted by ∂G. It may consist of several curves together with isolated points (see Figure 1.11).*

Definition (Cont.)

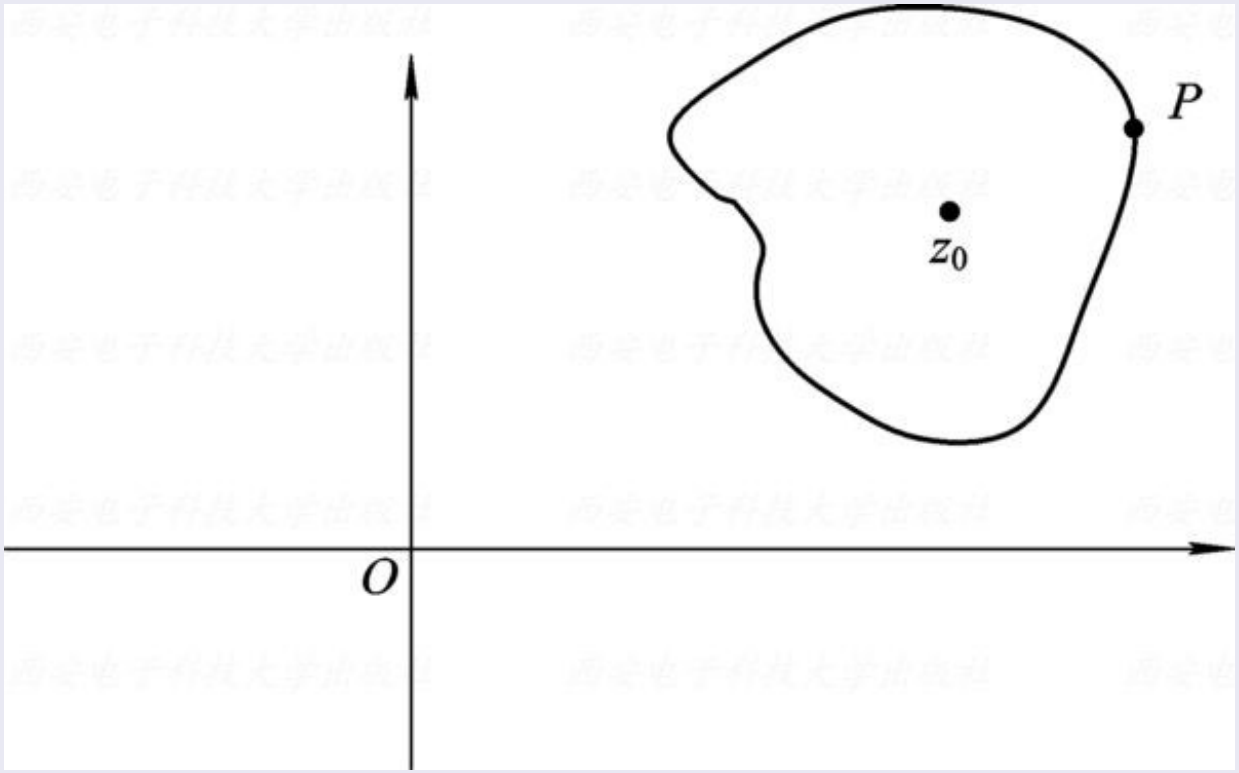


Figure: *1.11*

Definition

A point $z_0 \in G$ is an isolated point（孤立点） of G if there is a $\delta > 0$ such that $N_\delta^*(z_0) \cap G = \varnothing$.

Definition

A set G is bounded（有界） if there is an $R > 0$ such that $|z| < R$ for every $z \in G$. Otherwise, G is unbounded（无界）.

Definition

In the extended complex plane, a neighborhood of infinity（无穷远点邻域）*has the form*

$$N_R(\infty) = \{z : |z| > R\} \cup \{\infty\}, \qquad R > 0.$$

Removing ∞ *gives the deleted neighborhood of infinity*（无穷远点去心邻域）$N_R^*(\infty) = \{z : |z| > R\}$.

2 The Concept of Domains

Definition

A nonempty subset D of the complex plane is a domain（区域） if it satisfies both conditions:

- *D is an open set;*
- *D is connected（连通）: any two points in D can be joined by a polygonal path（折线路径） lying entirely in D.*

Thus, a domain is an open connected set. Its closure $\overline{D} = D \cup \partial D$ is sometimes called the associated closed domain（闭区域）.

The following sets are domains:

1. The open disk（开圆盘）centered at the origin: $|z| < R$;
2. The annulus（圆环域）in Figure 1.12(a): $r < |z| < R$;
3. The horizontal strip（水平带状域）in Figure 1.12(b): $y_1 < \operatorname{Im} z < y_2$;
4. The upper and lower half-planes（半平面）: $\operatorname{Im} z > 0$ and $\operatorname{Im} z < 0$;
5. The left and right half-planes: $\operatorname{Re} z < 0$ and $\operatorname{Re} z > 0$;
6. The sector（角域）in Figure 1.12(c): $0 < \arg z < \varphi$, where $0 < \varphi \leq \pi$.

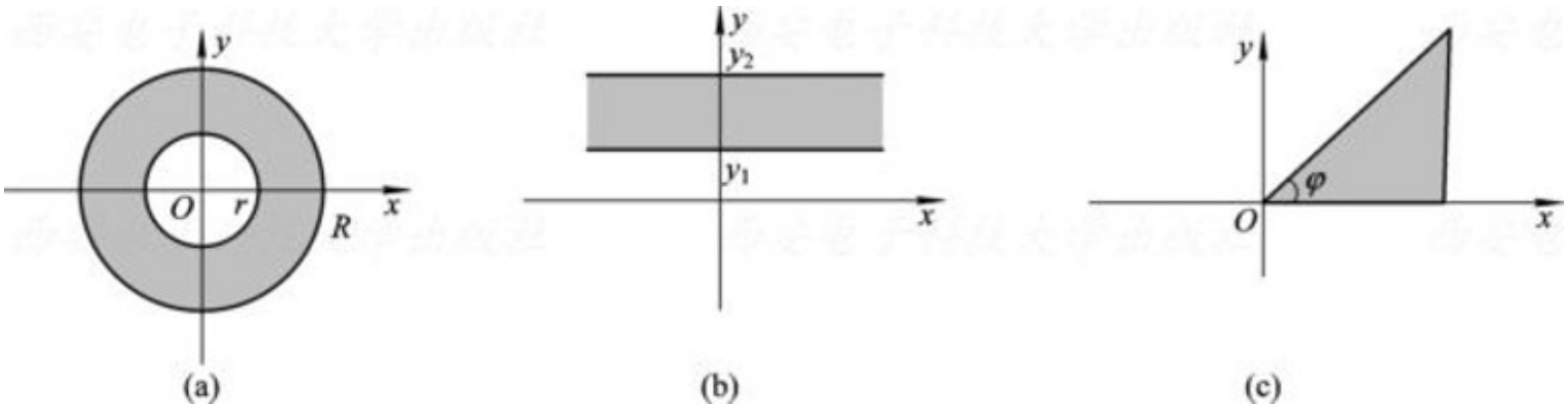


Figure: 1.12

3 Jordan Curve

Definition

Let $x(t)$ and $y(t)$ be continuous real-valued functions on $[a, b]$. The parametric equations

$$x = x(t), \qquad y = y(t) \quad (a \le t \le b)$$

define a continuous curve (连续曲线) in the plane. Equivalently, the curve has the complex parametrization

$$z = z(t) = x(t) + iy(t) \quad (a \le t \le b).$$

The points $z(a)$ and $z(b)$ are its initial point (起点) and terminal point (终点), respectively.

Definition (Cont.)

- *If $t_1 \neq t_2$ and $z(t_1) = z(t_2)$, the common point is called a self-intersection point(自交点)of C. For a closed curve, the starting point and the endpoint coincide, but this is not considered a self-intersection.*
- *A continuous curve with no self-intersection is called a simple curve*（简单曲线）.
- *If $z(a) = z(b)$ and there are no other self-intersections, then C is a simple closed curve, also called a Jordan curve*（若尔当曲线）.

For example, the circle $|z| = R$ is a Jordan curve. It separates the complex plane into two disjoint domains: the bounded interior $|z| < R$ and the unbounded exterior $|z| > R$. Their common boundary is $|z| = R$.

Definition

Let $C : z = z(t) = x(t) + iy(t)$, $\alpha \le t \le \beta$, *be a simple or simple closed curve. If* $x'(t)$ *and* $y'(t)$ *are continuous and*

$$[x'(t)]^2 + [y'(t)]^2 \neq 0$$

for every t, *then* C *is a smooth curve* (光滑曲线) . *A curve made of finitely many smooth arcs is a piecewise smooth curve* (分段光滑曲线) .

4 Simply and Multiply Connected Domains

Simple closed curves distinguish simply connected domains from multiply connected domains.

Definition

A domain D is simply connected (单连通) *if the interior of every simple closed curve contained in D also lies in D. Otherwise, D is multiply connected* (多连通) *.*

For example, the disk $|z| < R$ is simply connected, whereas the punctured disk (穿孔圆盘) $0 < |z| < R$ is multiply connected because it has a hole at $z = 0$. More generally, a domain containing a hole is multiply connected.

Example 3.1

Describe the point set $P = \{z = r(\cos\theta + i\sin\theta)\}$ *for which*

$$2\cos\theta < r < 4\cos\theta \left(-\frac{\pi}{2} < \theta < \frac{\pi}{2}\right).$$

Is this set a domain? If so, is it bounded or unbounded, and is it simply or multiply connected?

Solution.

Write

$$z = x + iy = r(\cos\theta + i\sin\theta)$$

Then

$$r = \sqrt{x^2 + y^2}$$

$$\cos\theta = \frac{x}{\sqrt{x^2 + y^2}}$$

Solution (Cont.)

Because $2\cos\theta < r < 4\cos\theta$,

$$2r\cos\theta < r^2 < 4r\cos\theta, \quad 2x < x^2 + y^2 < 4x$$

The first inequality gives $(x-1)^2 + y^2 > 1$, and the second gives $(x-2)^2 + y^2 < 4$. Thus, the point set is the shaded region in Figure 1.13:

$$\begin{cases} (x-1)^2 + y^2 > 1 \\ (x-2)^2 + y^2 < 4 \end{cases}$$

Therefore, the set is a bounded, simply connected domain.

Solution (Cont.)

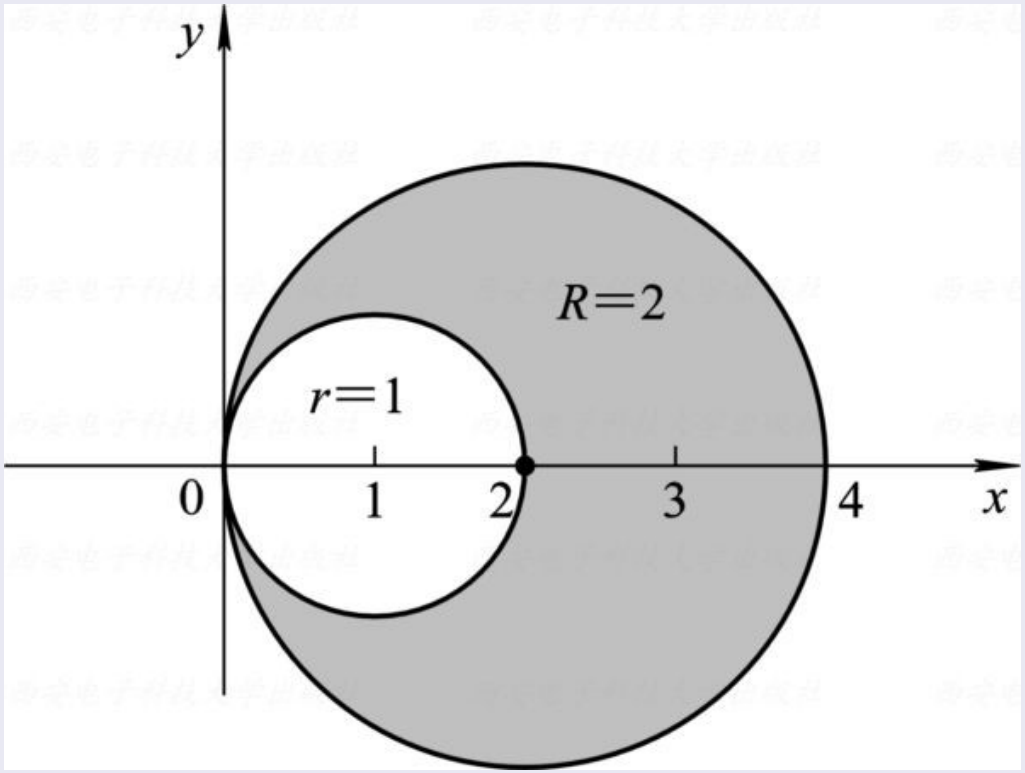


Figure: 1.13

Example 3.2

Does the set of points satisfying $\operatorname{Im} z > 1$ *and* $|z| < 2$ *form a domain? If so, is it bounded, and is it simply or multiply connected?*

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Does the set of points satisfying $\operatorname{Im} z > 1$ *and* $|z| < 2$ *form a domain? If so, is it bounded, and is it simply or multiply connected?*

Solution.

Let $z = x + iy$. The conditions become $y > 1$ and $x^2 + y^2 < 4$. The shaded set in Figure 1.14 is a bounded, simply connected domain.

Solution (Cont.)

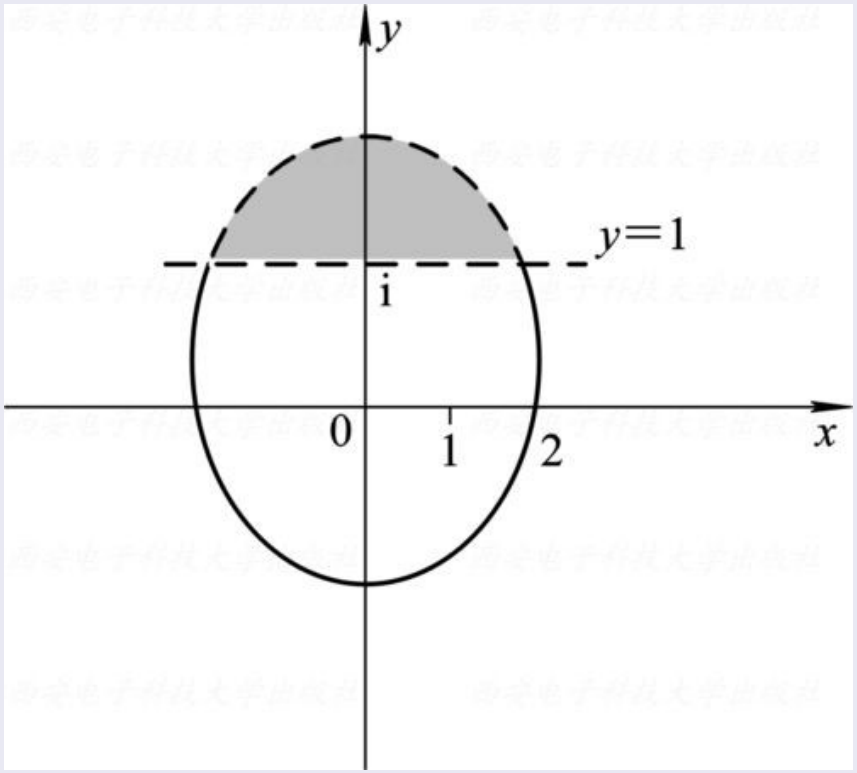


Figure: 1.14

Outline

1. Complex Numbers and Their Expressions
2. Operations of Complex Numbers
3. Domains in the Complex Plane
4. Complex Functions
5. Limits and Continuity of Complex Functions

1 Definition of a Complex Function

Definition

Let $G \subseteq \mathbb{C}$. A complex function（复变函数） f on G is a rule that assigns a unique complex number w to each $z \in G$. We write

$$w = f(z), \qquad z \in G.$$

The set G is the domain of definition（定义域） of f.

Definition (Cont.)

A function assigns exactly one value w to each z and is therefore single-valued（单值）*. A rule that assigns two or more values to some z is more precisely a multiple-valued correspondence*（多值对应）*, although it is often called a multiple-valued function. The set*

$$f(G) = \{f(z) : z \in G\}$$

is the range（值域）*of f.*

Definition (Cont.)

Writing $z = x + iy$ and $w = u + iv$ identifies each complex number with two real coordinates. Hence, a complex function $w = f(z)$ can be written as

$$f(x + iy) = u(x, y) + iv(x, y),$$

where u and v are real-valued functions (实值函数) of the two real variables x and y.

Definition (Cont.)

Let $w = f(z)$, *with* $z = x + iy$ *and* $w = u + iv$. *Since* $z \longleftrightarrow (x, y)$,

$$\begin{aligned} f(z) &= w = u + iv, \\ f(x+iy) &= \underbrace{\operatorname{Re} f(x+iy)}_{u(x,y)} + i \underbrace{\operatorname{Im} f(x+iy)}_{v(x,y)} \\ &= u(x, y) + iv(x, y). \end{aligned}$$

Definition (Cont.)

Conversely, any real-valued $u(x, y)$ and $v(x, y)$ on G define a complex function. Hence

$$w = f(z) \quad \Longleftrightarrow \quad \begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases} \qquad (x, y) \longmapsto (u, v).$$

The formula may involve $\overline{z}$, $|z|$, $\operatorname{Re} z$, x, y, not only z:

$$f(z) = \overline{z} = x - iy \quad \Longrightarrow \quad u(x, y) = x, \quad v(x, y) = -y.$$

For example, consider $w = z^2$. Let $z = x + iy$ and $w = u + iv$. Then

$$u + iv = (x + iy)^2 = x^2 - y^2 + 2xyi,$$

so $w = z^2$ corresponds to the two real-valued functions

$$u(x, y) = x^2 - y^2, \qquad v(x, y) = 2xy.$$

2 The Concept of Mapping

Under the mapping (映射) $w = f(z)$, the point w is the image (像) of z, and z is a preimage (原像) of w. We call the domain plane the z-plane and the image plane the w-plane (see Figure 1.15).

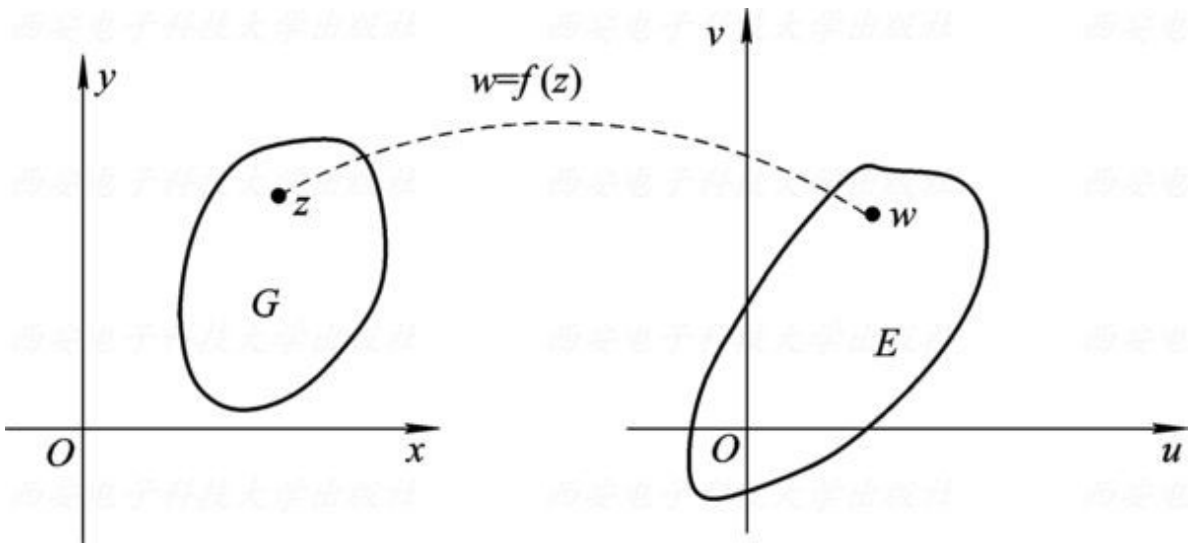


Figure: 1.15

For example, $w = \bar{z}$ maps $z = x+iy$ in the z-plane to $w = x-iy$ in the w-plane. Thus, $z_1 = 1+4i$ maps to $w_1 = 1-4i$, $z_2 = 1-3i$ maps to $w_2 = 1+3i$, and $\triangle ABC$ maps to $\triangle A'B'C'$ (see Figure 1.16).

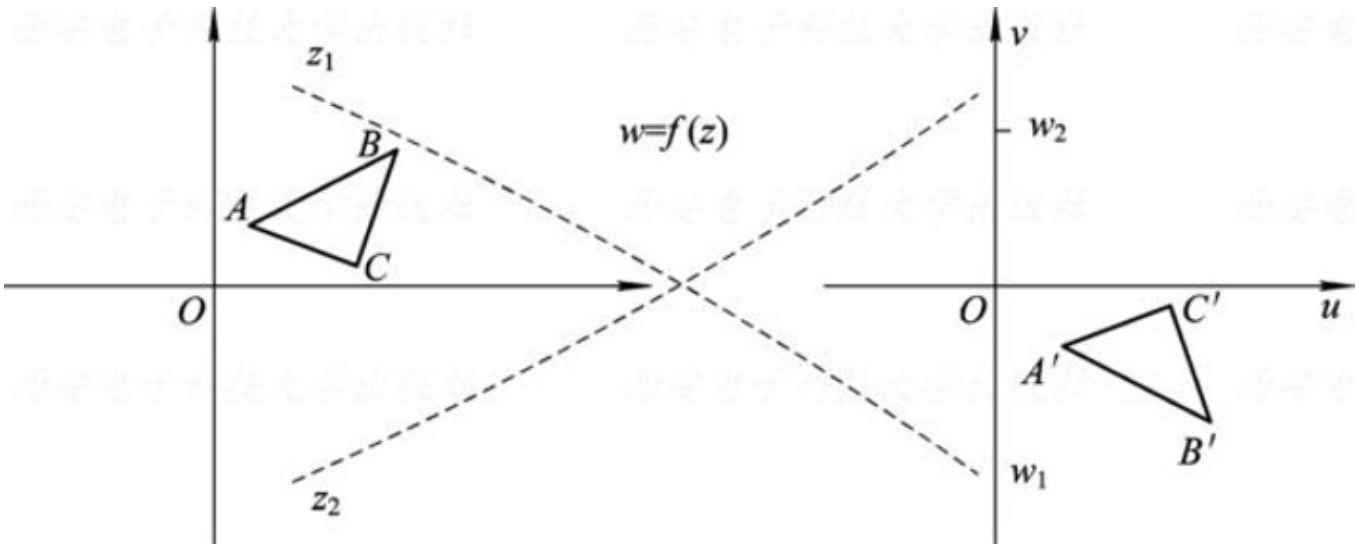


Figure: 1.16

Example 4.1

Investigate the mapping properties of the function $w = z^2$.

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Investigate the mapping properties of the function $w = z^2$.

Solution.

1. If $z = re^{i\theta}$ and $w = \rho e^{i\varphi}$, then $w = z^2$ gives

$$\begin{cases} \rho = r^2 \\ \varphi \equiv 2\theta \pmod{2\pi} \end{cases}$$

This yields the following mapping properties:

Solution (Cont.)

(1) The ray (射线) $\{re^{i\theta_0} : r \geq 0\}$ in the z-plane maps to $\{\rho e^{i2\theta_0} : \rho \geq 0\}$ in the w-plane, as shown in Figure 1.17(a), (b).

(2) For $0 \leq \alpha \leq \pi/2$, the sector $D = \{re^{i\theta} : r \geq 0,\ 0 \leq \theta \leq \alpha\}$ maps to $G = \{\rho e^{i\varphi} : \rho \geq 0,\ 0 \leq \varphi \leq 2\alpha\}$, as shown in Figure 1.17(c), (d).

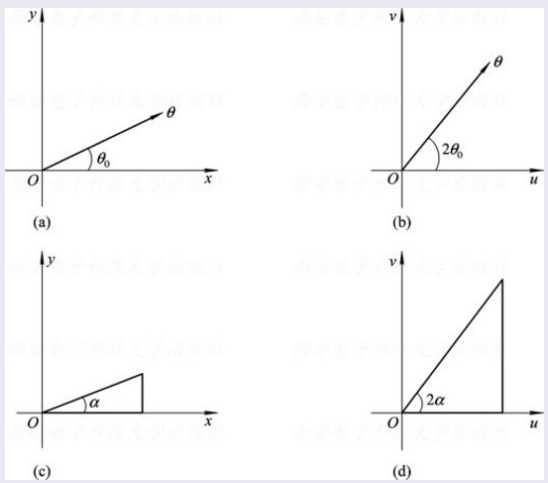


Figure: 1.17

Solution (Cont.)

2. If $z = x + iy$ and $w = u + iv$, then $w = z^2$ gives

$$\begin{cases} u = x^2 - y^2 \\ v = 2xy. \end{cases}$$

Therefore:

(1) For $c_1 \neq 0$, the curves $x^2 - y^2 = c_1$ are rectangular hyperbolas (等轴双曲线) with asymptotes (渐近线) $y = \pm x$; they map to the vertical lines $u = c_1$. For $c_2 \neq 0$, the curves $2xy = c_2$ have the coordinate axes as asymptotes; they map to the horizontal lines $v = c_2$ (see Figure 1.18).

Solution (Cont.)

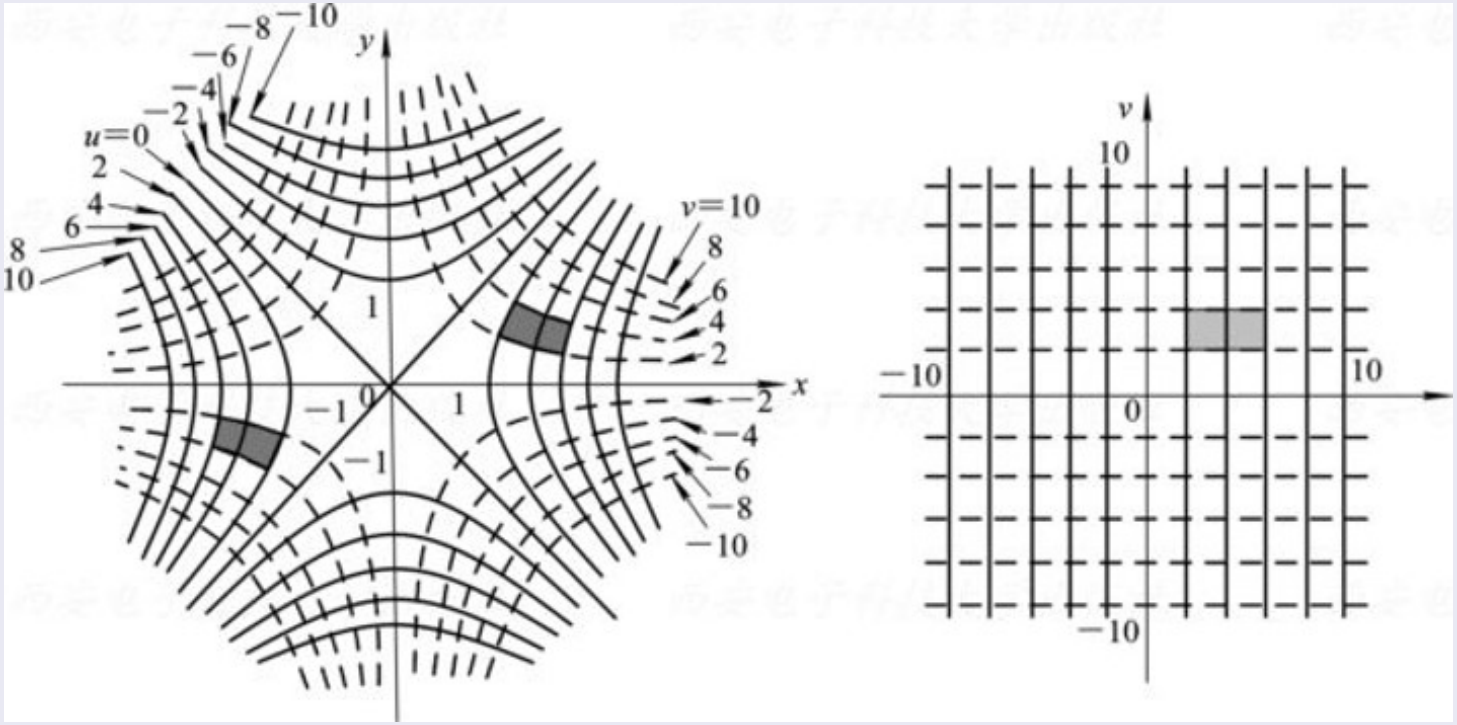


Figure: 1.18

Solution (Cont.)

2 We next determine the images of the two families of lines $x = c_1$ and $y = c_2$ under $w = z^2$.
For $x = c_1$, a line parallel to the imaginary axis in the z-plane, we have

$$\begin{cases} u = c_1^2 - y^2 \\ v = 2c_1 y \end{cases} \quad (-\infty < y < \infty).$$

Eliminating y gives

$$v^2 = 4c_1^2(c_1^2 - u)$$

Solution (Cont.)

These are parabolas (抛物线) in the w-plane, shown as dashed curves in Figure 1.19(b).
When $c_1 = 0$, the line is the imaginary axis $x = 0$, and its image satisfies

$$\begin{cases} u = -y^2 \\ v = 0 \end{cases} \Rightarrow \begin{cases} u \leq 0 \\ v = 0 \end{cases}$$

Thus, $w = z^2$ maps the imaginary axis in the z-plane onto the nonpositive real axis in the w-plane.

Solution (Cont.)

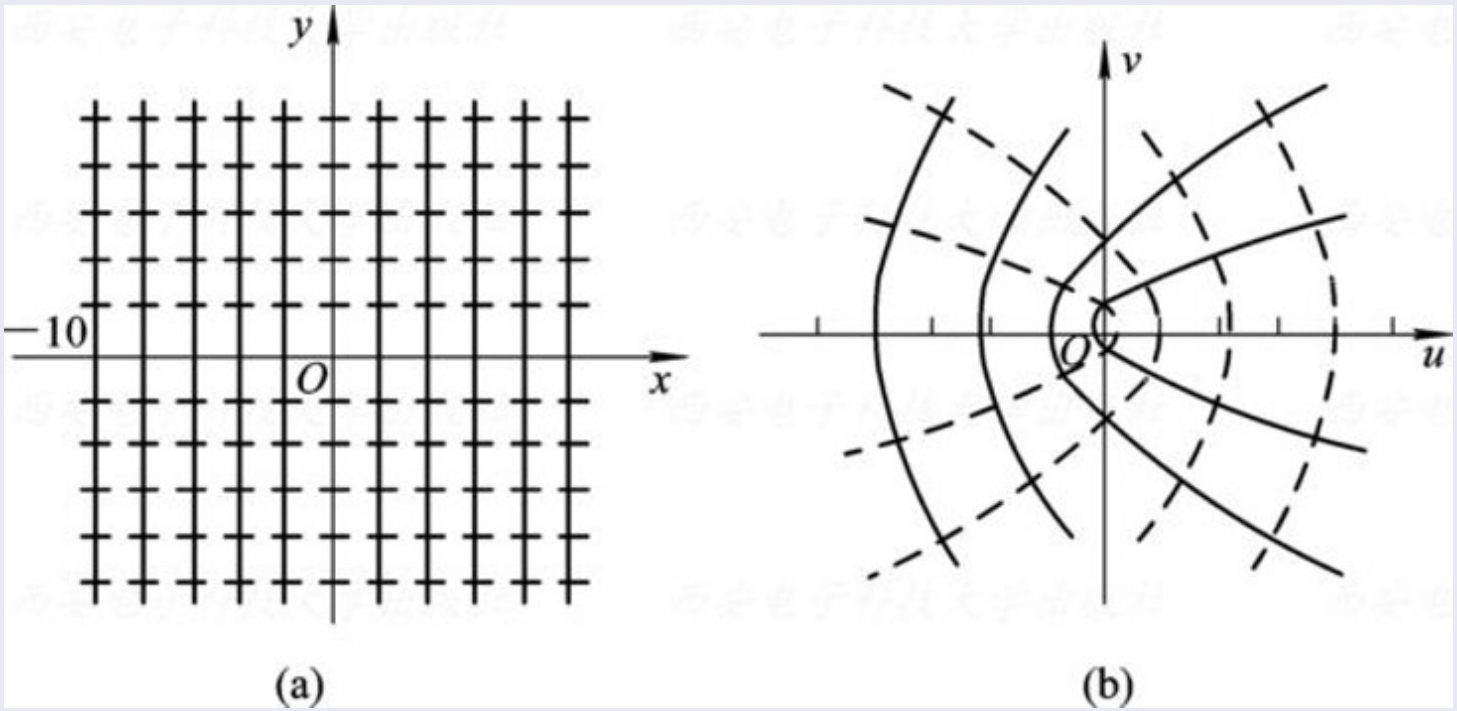


Figure: 1.19

Solution (Cont.)

Similarly, the line $y = c_2$, shown dotted in Figure 1.19(a), maps under $w = z^2$ to the parabola

$$v^2 = 4c_2^2\left(u + c_2^2\right)$$

These are the solid curves in Figure 1.19(b). When $c_2 = 0$, the line is the real axis $y = 0$, and its image satisfies

$$\begin{cases} u = x^2 \\ v = 0 \end{cases} \text{ that is } \begin{cases} u \geq 0 \\ v = 0 \end{cases}$$

Thus, the real axis in the z-plane maps onto the nonnegative real axis in the w-plane.

Example 4.2

Under the mapping $w = z + \dfrac{1}{z}$, *find the image of the circle* $|z| = 2$.

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Solution.

Let $z = x + iy$ and $w = u + iv$. Then

$$u + iv = x + iy + \frac{x - iy}{x^2 + y^2},$$

so

$$\begin{cases} u = x + \dfrac{x}{x^2 + y^2} \\ v = y - \dfrac{y}{x^2 + y^2} \end{cases} \tag{1}$$

Solution (Cont.)

Parametrize $|z| = 2$ by $x = 2\cos\theta$ and $y = 2\sin\theta$, where $0 \le \theta \le 2\pi$. Substitution into (1) gives

$$u = \frac{5}{2}\cos\theta, \qquad v = \frac{3}{2}\sin\theta.$$

Therefore, the image is the ellipse (椭圆)

$$\frac{u^2}{(5/2)^2} + \frac{v^2}{(3/2)^2} = 1$$

in the w-plane.

Outline

1. Complex Numbers and Their Expressions
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1 Limits of Complex Functions

Definition

Suppose f is defined in a deleted neighborhood of z_0. A complex number A is the limit (極限) of $f(z)$ as $z \to z_0$ if, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$0 < |z - z_0| < \delta \quad \Longrightarrow \quad |f(z) - A| < \varepsilon.$$

We write

$$\lim_{z \to z_0} f(z) = A$$

or $f(z) \to A$ as $z \to z_0$ (see Figure 1.20).
The same value A must be obtained along every path by which z approaches z_0.

Definition (Cont.)

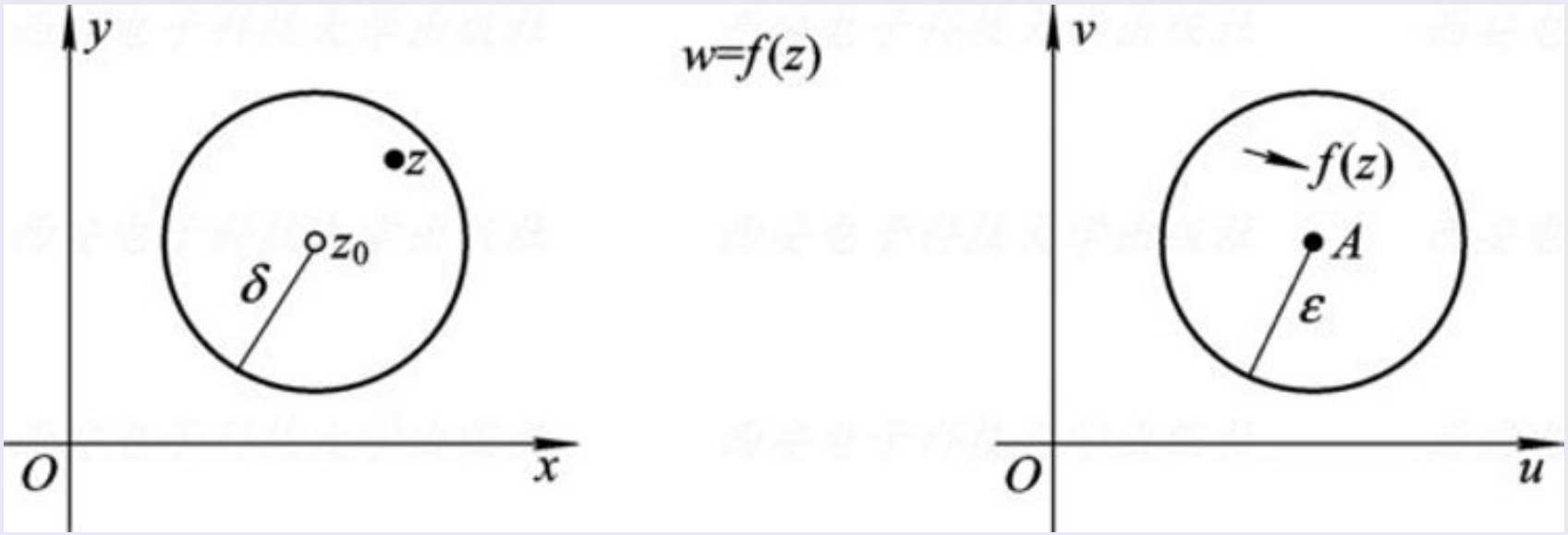


Figure: *1.20*

Theorem 5.1

Let $f(z) = u(x, y) + iv(x, y)$, $A = a + ib$, *and* $z_0 = x_0 + iy_0$. *Then*

$$\lim_{z \to z_0} f(z) = A$$

if and only if

$$\lim_{\substack{x \to x_0 \\ y \to y_0}} u(x, y) = a, \quad \lim_{\substack{x \to x_0 \\ y \to y_0}} v(x, y) = b.$$

Proof.

Assume $\lim_{z \to z_0} f(z) = A$. For every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$$

implies

$$|(u + iv) - (a + ib)| < \varepsilon$$

that is,

$$|(u - a) + i(v - b)| < \varepsilon.$$

Proof (Cont.)

Since the absolute values of the real and imaginary parts do not exceed the modulus,

$$|u - a| < \varepsilon, \quad |v - b| < \varepsilon$$

whenever $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$. Therefore,

$$\lim_{\substack{x \to x_0 \\ y \to y_0}} u(x, y) = a, \quad \lim_{\substack{x \to x_0 \\ y \to y_0}} v(x, y) = b$$

Proof (Cont.)

Conversely, assume the two real limits hold. Given $\varepsilon > 0$, choose $\delta_1, \delta_2 > 0$ such that

$$|u - a| < \frac{\varepsilon}{2} \quad \text{if } 0 < |z - z_0| < \delta_1,$$

$$|v - b| < \frac{\varepsilon}{2} \quad \text{if } 0 < |z - z_0| < \delta_2.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. Then

$$|f(z) - A| = |(u - a) + i(v - b)| \leq |u - a| + |v - b|$$

so $0 < |z - z_0| < \delta$ implies $|f(z) - A| < \varepsilon$. Hence, $\lim_{z \to z_0} f(z) = A$.

Property 5.1

If $\lim_{z \to z_0} f(z)$ *exists, it is unique.*

Property 5.2

If $\lim_{z \to z_0} f(z)$ *exists, then* f *is bounded in some deleted neighborhood of* z_0.

Intuition: Uniqueness of the Limit

Suppose $f(z) \to A$ and $f(z) \to B$ with $A \neq B$. Take

$$\varepsilon = \frac{|A-B|}{2}: \qquad \{w : |w-A| < \varepsilon\} \cap \{w : |w-B| < \varepsilon\} = \varnothing.$$

For z close enough to z_0, $f(z)$ would lie in both disks:

$$|A-B| \leq |A-f(z)| + |f(z)-B| < \varepsilon + \varepsilon = |A-B|,$$

a contradiction. Hence $A = B$.

Intuition: $f(z)$ cannot approach two different points at the same time.

Intuition: Local Boundedness

Take $\varepsilon = 1$. There is a $\delta > 0$ such that

$$\begin{aligned} 0 < |z - z_0| < \delta &\implies |f(z) - A| < 1 \\ &\implies |f(z)| \le |f(z) - A| + |A| < |A| + 1. \end{aligned}$$

Intuition: $f(z)$ is tied to A by a string of length 1.

- The bound is only local: $f(z) = \dfrac{1}{z}$ satisfies $\lim\limits_{z \to 1} f(z) = 1$ and is bounded near 1, but it is unbounded near 0.
- The value $f(z_0)$ plays no role, because the neighborhood is deleted.
- Contrapositive: if f is unbounded near z_0, e.g. $\dfrac{1}{z}$ near 0, then $\lim\limits_{z \to z_0} f(z)$ does not exist.

Theorem 5.2

Suppose f and g satisfy

$$\lim_{z\to z_0} f(z) = A, \quad \lim_{z\to z_0} g(z) = B.$$

Then

1. $\lim_{z\to z_0} [f(z) \pm g(z)] = A \pm B;$
2. $\lim_{z\to z_0} [f(z)g(z)] = AB;$
3. $\lim_{z\to z_0} \dfrac{f(z)}{g(z)} = \dfrac{A}{B} \quad (B \neq 0).$

Example 5.1

Show that $f(z) = \dfrac{\operatorname{Re} z}{|z|}$ *has no limit as* $z \to 0$.

Proof.

For $z = x + iy$,

$$u(x, y) = \frac{x}{\sqrt{x^2 + y^2}}, \qquad v(x, y) = 0$$

Proof (Cont.)

Along the positive real axis, let $z = x$ with $x \to 0^+$. Then

$$f(x) = \frac{x}{|x|} = 1.$$

Along the imaginary axis, let $z = iy$ with $y \to 0$. Then

$$f(iy) = 0.$$

The two path limits are different, so $\lim_{z \to 0} f(z)$ does not exist. □

Example 5.2

Find the following limits:

1. $\lim\limits_{z\to i} \dfrac{z-i}{z(1+z^2)}$
2. $\lim\limits_{z\to 1} \dfrac{z\bar{z}+2z-\bar{z}-2}{z^2-1}$

Solution.

1

$$\lim_{z\to i}\frac{z-i}{z(1+z^2)}=\lim_{z\to i}\frac{z-i}{z(z-i)(z+i)}=\lim_{z\to i}\frac{1}{z(z+i)}=-\frac{1}{2}$$

2

$$\lim_{z\to 1}\frac{z\bar{z}+2z-\bar{z}-2}{z^2-1}=\lim_{z\to 1}\frac{(\bar{z}+2)(z-1)}{(z+1)(z-1)}=\lim_{z\to 1}\frac{(\bar{z}+2)}{(z+1)}=\frac{3}{2}$$

2 Continuity（连续性）of Complex Functions

Definition

Let f be defined near z_0. It is continuous at z_0 if

$$\lim_{z \to z_0} f(z) = f(z_0).$$

Continuity at ∞ is defined analogously using neighborhoods of infinity. If f is continuous at every point of a domain D, then f is continuous on D.

Theorem 5.3

The complex function $f(z) = u(x, y) + iv(x, y)$ *is continuous at* $z_0 = x_0 + iy_0$ *if and only if both* u *and* v *are continuous at* (x_0, y_0)*, that is,*

$$\lim_{\substack{x \to x_0 \\ y \to y_0}} u(x, y) = u(x_0, y_0)$$

$$\lim_{\substack{x \to x_0 \\ y \to y_0}} v(x, y) = v(x_0, y_0)$$

Theorem 5.4

1. *Sums, differences, and products of functions continuous at z_0 are continuous at z_0. Their quotient is also continuous there if the denominator is nonzero at z_0.*
2. *If g is continuous at z_0 and f is continuous at $g(z_0)$, then the composite function (複合函数) $f \circ g$ is continuous at z_0.*

Example 5.3

Prove that $f(z)$ is not continuous at the origin, where

$$f(z) = \begin{cases} \dfrac{xy}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

Proof.

If f were continuous at the origin, then

$$\lim_{z \to 0} f(z) = f(0) = 0.$$

Proof (Cont.)

Along the line $y = x$, let $z = x + ix$ with $x \to 0$. Then

$$f(x + ix) = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Along the imaginary axis, let $z = iy$ with $y \to 0$. Then

$$f(iy) = 0.$$

The path limits differ, so the limit does not exist. Hence, f is not continuous at the origin. □

A function f defined on a curve C is continuous at $z_0 \in C$ if

$$\lim_{z \to z_0} f(z) = f(z_0) \quad (z \in C)$$

where z is restricted to C. If C is parametrized over a closed bounded interval and f is continuous on C, then f is bounded on C. Thus, there exists $M > 0$ such that

$$|f(z)| \leq M \qquad (z \in C).$$

Intuition: Continuity along a Curve

Along C, z may approach z_0 only through points of C. For example,

$$f(z) = \begin{cases} \dfrac{xy}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases} \qquad C: \ z(t) = t, \ t \in [-1, 1].$$

- In the plane: f is not continuous at 0 (along $y = x$ the limit is $1/2$).
- On C: $y = 0$, so $f(z) \equiv 0$ and f is continuous on C.

Restricting z to C leaves only one way to approach z_0, so the condition is weaker.

Intuition: Why a Closed Bounded Parameter Interval?

Parametrization turns the curve C into an interval $[a, b]$:

$$z = z(t),\ t \in [a, b] \implies g(t) = |f(z(t))| \text{ continuous on } [a, b].$$

By the extreme value theorem（最值定理）, $g(t) \le g(t^*) = M$ for some $t^* \in [a, b]$.

- Not closed: $z(t) = t,\ t \in (0, 1],\ f(z) = 1/z$ is continuous but unbounded.
- Not bounded: $z(t) = t,\ t \in [1, +\infty),\ f(z) = z$ is continuous but unbounded.

In Chapter 3, this M gives the estimate $\left|\int_C f\,dz\right| \le ML$.